**ClickU:**

**A Web-based Kitchen Equipment Circulation and Inventory System**

A Proposal

Presented to the Faculty of the

College of Computer Studies, University of Cebu Lapu-Lapu and Mandaue

Mandaue City, Cebu

In Partial Fulfillment of the Requirements

For the Degree of Bachelor of Science in Information Technology

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# DEDICATION

"A dream does not become reality through magic; it takes sweat, determination and hard work."

-Colin Powell

We the researchers would like to dedicate this work to the following that has always believe in us, trusted our genuine hard-work, always been there with us, one call away, and continuously help us develop our potentials and skills as we are striving for our success in this journey:

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CCS Family
Dear Friends
&
The Almighty God

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The Researchers

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# CHAPTER I
# INTRODUCTION

## Rationale of the Study

Technology is far and wide. Look around you and observe how many things is technology. There may be telephone, television, radio, and etcetera. Technology refers to systems, and devices are the results of scientific knowledge base on being used for practical and daily purposes (Definition of Technology, 2019). It is the source of many environmental and social problems. It is used to accomplish several tasks in daily life. It is practiced in almost everything people do in their daily lives. It is used at work, at school and everywhere it is used for communication, transportation, learning, securing data, purchasing and so much more. Technological improvements are refining the world exceedingly and play a crucial role in molding and maturing human society. Through these growth of technology with the integration of web technology and various mobile and software applications, tasks are done more systematically and more efficiently. One of the technological advancement is the web-based system.

Web-based system provides access to system software using computer and Intranet connection. It is a software environment accessible via Web browser, containing an integrated set of tools and functionalities. Additionally, it is one of the advancements of the technology that people face now a day. Web-based systems have evolved compellingly over recent years and with enhancement in security and technology. A web-based system is any system that uses a website as the interface or front-end. Users can easily access the system from any computer connected to the Intranet using a standard browser. Automated form is one of the examples of many web-based systems.

Paper is an important material used daily for many purposes worldwide since then. Take a look around and notice how many things are made out of paper. There may be books, magazines, tissues, newspapers and notebooks. The world produces 300 million tons of paper. According to the U.S. Environmental Protection Agency, printing and writing papers usually found in a very college or workplace setting like duplicator paper, laptop printouts and notepads, comprise the most important class of paper product consumption and landfill waste. This is an alarming figure considering the advances technology of today’s world that is specifically designed to help offices go paperless (O'Mara, 2019).

The Hospitability Management (HM) Department of University of Cebu Lapu-Lapu and Mandaue are still using papers for their daily transactions of their kitchen and other laboratories.

Although this method is still effective, there are still disadvantages to it. One problem on using this current method is that these papers can be easily lost. It is also so hard to track transactions or reports. Saving, keeping or tracking all those paper transactions is another problem. Borrowing the utensils also adds too much time in filling up the forms. The assigned staffs in each laboratory are having the same concerns that they suggested the process to be real-time and more efficient.

The current method has many flaws but imagine this method being upgraded with technology. Imagine no more hassle in filling up forms just to borrow the utensils. Imagine no more scanning all through the files just to find that one paper that is needed. Imagine doing all this through a computer or through a mobile. TRES n' PEACE would like to make this possible by creating a system that would cover all of this. Making the process a hassle-free not just for the borrowers but also for the faculty staffs.

## Objective of the Study

### General Objective

The main objective of this study is to develop a web-based Kitchen Equipment Circulation and Inventory System for Hospitability Management Department of University of Cebu Lapu-Lapu and Mandaue.

### Specific Objectives

Specifically, this aims to:

1. Evaluate the existing inventory management currently in place in the department;
2. Identify problem areas encountered in the existing process used by the department.
3. Document into details the computerization process of the proposed system.

## Scope and Limitation of the Study

The study focuses on developing a web-based system for Kitchen Equipment Circulation and Inventory System. The main scope of the system is identified and described below.

*User Account Registration Module.* This function provides the admins to create, manage and login in their accounts. By using this module the department dean can view the time-logs of kitchen staff and the lab-in-charge.

*Reservation of Equipment Process Module.* Reservation of equipment process provides access to information for the borrower. This module provides the borrower that enables them to view the list of all available kitchen equipment with their corresponding quantity. The borrower can search the kitchen equipment, can reserve and can cancel their requested kitchen equipment; the approval of the reservation will be from the kitchen staff.

*Borrowing of Equipment Process Module.* This includes the borrowing and approval of the kitchen equipment that is requested by the borrower. The approval of the request will be from the kitchen staff. This module includes the cancellation of the requested kitchen equipment from the borrower.

*Returning of Equipment Process Module.* This includes the returning and status of the kitchen equipment that is returned by the borrower; the status of kitchen equipment will be from the kitchen staff. This module includes the replacement of the loss kitchen equipment.

*Inventory System Module.* Inventory System is a systematize process that provides the quantity of kitchen equipment that is available, borrowed, returned and the original quantity and the current quantity. This also monitors the utilization of the kitchen equipment. It also includes the adding, updating, restocking, tracking and deleting of kitchen equipment to the general storage.

*Generate Report Module.* This function provides access to information for the admins. This module provides the admins that enables them to view all transaction like borrowed logs, returned logs, pending transaction and summary of kitchen equipment from kitchen laboratory and general storage.

However, the proposed system does not cover purchasing of equipment and packaging processes and monitoring and computation of overdue fines. The application accessibility will require internet connectivity. The system data will be stored in the local server by means of Intranet.

## Significance of the Study

This study is conducted for the benefits of the following:

*Borrower*. The study will help the borrower to know if they have a pending transaction and/or unreturned kitchen equipment. Fast and easy access of the system and the borrower will no longer fill up borrower sheets that consumes too much of their time.

*Kitchen Staff*. The study will help the kitchen staff to minimize workload and maximize work.

*Gen. Storage In charge*. The study will help the lab-in-charge to monitor the kitchen equipment in paperless way.

*Department Dean*. The study will help the department dean to monitor logs as well as the kitchen equipment.

*Researchers*. The study will be able to give an opportunity to the researchers to resolve complex methods and enhance programming skills.

*Future Researchers*. This will help future researchers to improve the system in the near future.

**Flow of the Study**

The figure below shows the three processing tasks which are the Input, Process and the Output of the proposed system. In the first phase, the flow shows the defined problem that will be the focus of the researchers. On the second phase, it reveals the data gathering and interpretation. On the third phase, it exposes the expected output of the research, the ClickU: Kitchen Equipment Circulation and Inventory System.

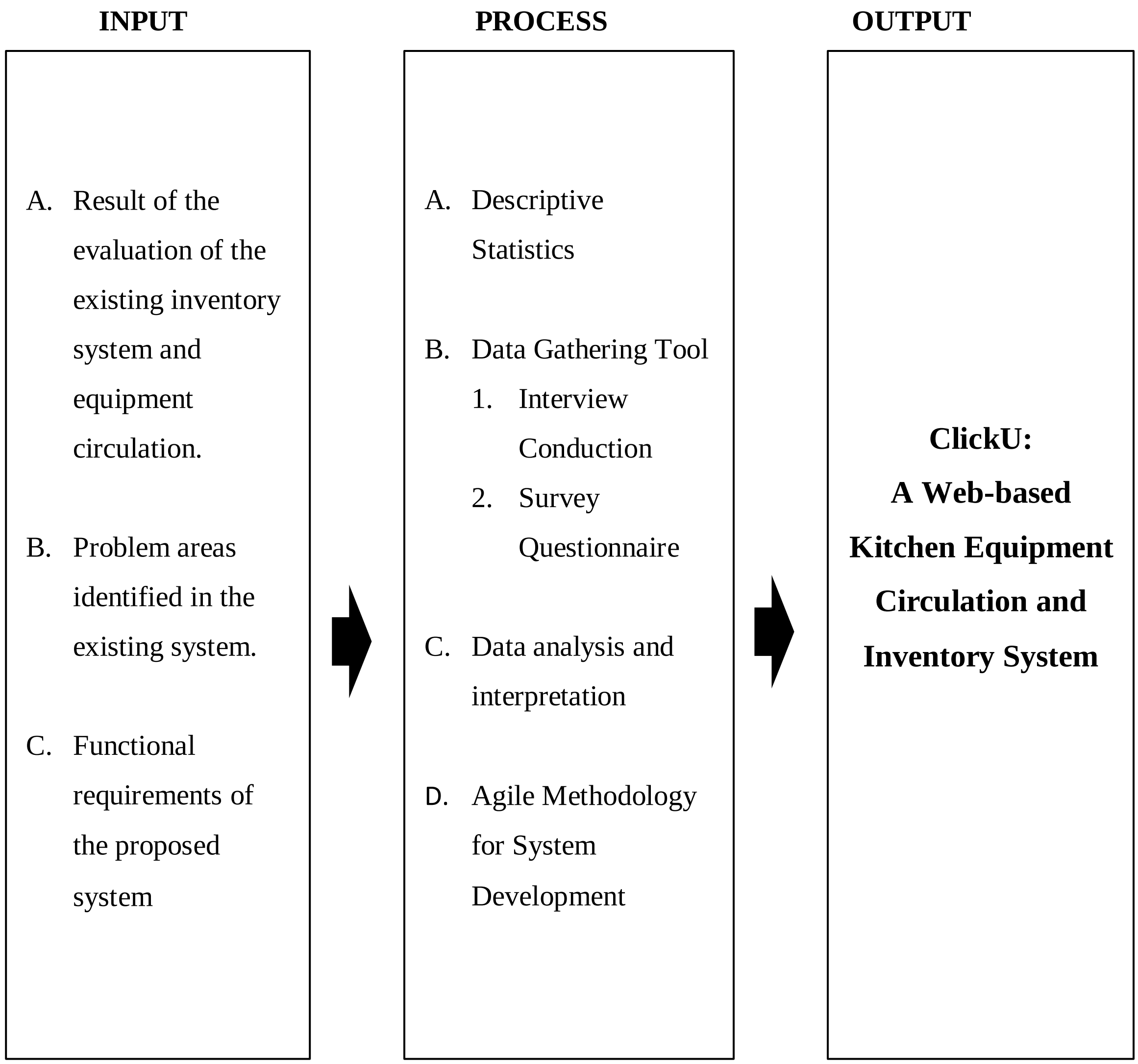



Figure 1: **Flow of the Study of ClickU**

Figure 1 shows the flow of the study which is the three processing tasks which are input, process and output of ClickU.

## Definition of Terms

**Agile Methodology** is an umbrella term for a set of methods and practices based on the values and principles expressed in the Agile Manifesto.

**Borrower** is the one who will utilize the system and is the primary prospect of the system and who will use the kitchen equipment and the kitchen laboratory.

**ClickU** is a Web-based Kitchen Equipment Circulation and Inventory System of kitchen equipment for the HM Department of UCLM.

**Dean** refers to the admin which is the department dean of HM Department

**General Storage** is where all the kitchen equipment is stored.

**HM Department** is what the majority of user's course and also where the dean's department.

**KECIS** is a systematic process of borrowing, returning and inventory of kitchen equipment.

**Kitchen Laboratory** where the students will perform their activity like cooking.

**Kitchen Staff** is tasked at the laboratory.

**Gen. In charge** is assigned to monitor the general storage of kitchen equipment

**Server** is where the students will connect their mobile device in order to access the system and make transaction.

**Web-based System** is a software system using a computer and Intranet connection.

# CHAPTER II

# REVIEW OF RELATED LITERATURE AND STUDIES

To understand the study further, the researchers made use of different reading materials related to the proposed system. These materials such as books, magazines, and other web articles that is essential in broadening the knowledge of the researchers. These can guide the researchers to realize their target objectives by obtaining concepts in alternative connected studies and build enhancements as potential.

## Related Literature

A laboratory is a facility in university where students will perform their activities, experiment (Laboratory, 2019). The aim of laboratory work is to deepen and fix theoretical data and to develop the talents of freelance experimentation. The work includes making ready the equipment, equipment, associated reagents necessary for an experiment, diagraming and coming up with the experiment, closing the experiment itself and writing a laboratory report. Laboratory work is widely used in teaching the natural sciences with the appropriate relation of theoretical to laboratory work established for each and every field. As a rule, laboratory work ends with a final report on the complete cycle of experiments performed. There are lot of types of laboratories, for example speech laboratory, physics laboratory, chemistry laboratory, kitchen laboratory and many more.

A room laboratory may be a space or a part of an area used for preparation and food preparation. It is usually equipped with a stove, a sink with hot and cold running water, an icebox, and worktops and room cupboards organized in line with a standard style. The main functions are to store, prepare and cook food (Kitchen, 2019). Choosing the proper quite room instrumentation helps in creating preparation fun and simple. Kitchen equipment is entirely essential for preparing and serving food. The list of kitchen equipment incorporates all key equipment ranging from knives to huge boiler. Kitchen equipment is entirely essential for preparing and serving food. The list of kitchen equipment incorporates all key equipment ranging from knives to huge boiler. It is entirely essential to select utensils which will be used together with your kitchen utensils. A utensil may be a little handheld tool used for food preparation. Common room tasks embrace cutting food things to size, heating food on associate degree shoot or on a stove, baking, grinding, mixing, blending, and measuring; completely different utensils square measure created for each task (Kitchen utensil, 2019). This kitchen equipment is reserved, borrowed and returned by the borrower who uses the kitchen laboratory.

In reserving, borrowing and returning kitchen equipment the borrower must observe the proper etiquette of financing. Being a borrower bring a terrific amount of responsibility and must be extra cautious in conduct kitchen equipment. In borrowing kitchen equipment it must be approved by the kitchen staff. Borrower must return after using it as soon as possible or within the return period. In returning the kitchen equipment it must be complete and in good condition hence, if the borrower lost or damaged the kitchen equipment it must be repair or replaced directly if it cannot be fixed (Mayne, 2019).

## Related Studies

The information below shows the five different related studies of ClickU: Kitchen Equipment Circulation and Inventory System, namely: inFlow Inventory, Odoo, Sortly Pro, ZhenHub and Zoho Inventory.

### inFlow Inventory

inFlow Inventory suits businesses of all sizes. Its free version is deployed on-premise and enables to manage up to one hundred merchandise and customers. This version includes barcoding, price management, sales orders, purchase orders, and count sheet functionalities. inFlow's unique selling point is its payments tracking functionality, which provides real-time details of all the completed and pending payment transactions (Srivastava, 2019).

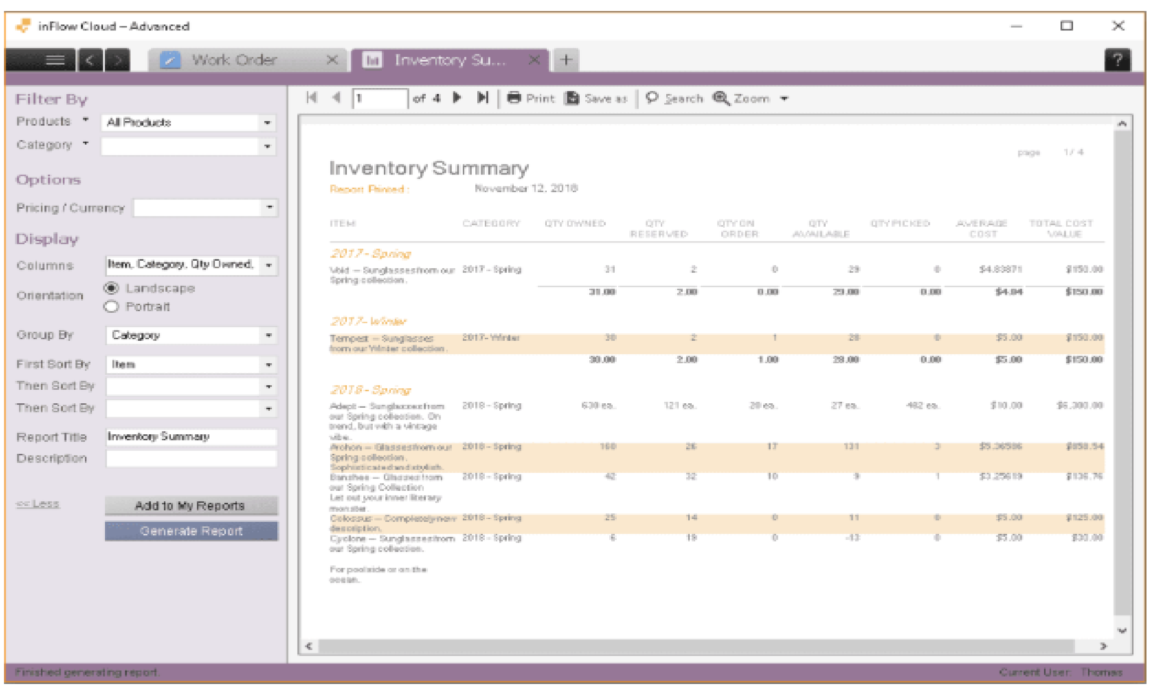



Figure 2 : **<u>inFlow Inventory</u>**

Figure 2 shows the Inventory Summary of inFlow Inventory.

**Odoo**

Odoo is an open source enterprise resource planning solution for businesses of all sizes. Users can download and access its inventory module, using it as a standalone inventory management solution. Odoo is offered at no cost if you implement solely the inventory management module (Srivastava, 2019).

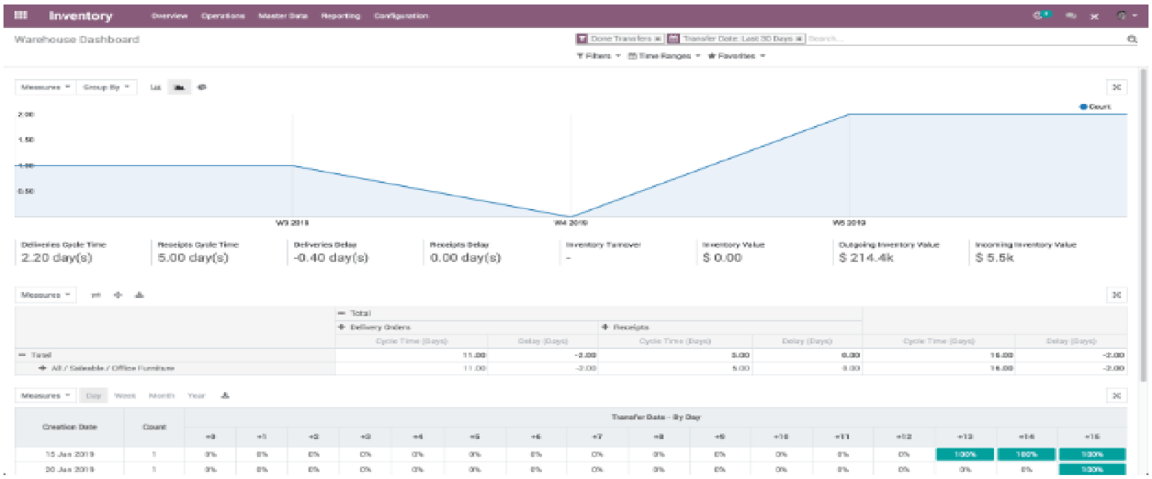



Figure 3 : **<u>Odoo</u>**

Figure 3 shows that the analysis of the data gathered by using Odoo.

**Sortly Pro**

Sortly Pro is a cloud-based inventory management solution for businesses of all sizes. It is free plan supports one user and helps to manage up to one hundred dealing transaction entries per month. Sortly Pro's unique selling point is its product tagging and cataloging functionality that lets users create product catalogs with up to eight photos for each item (Srivastava, 2019).

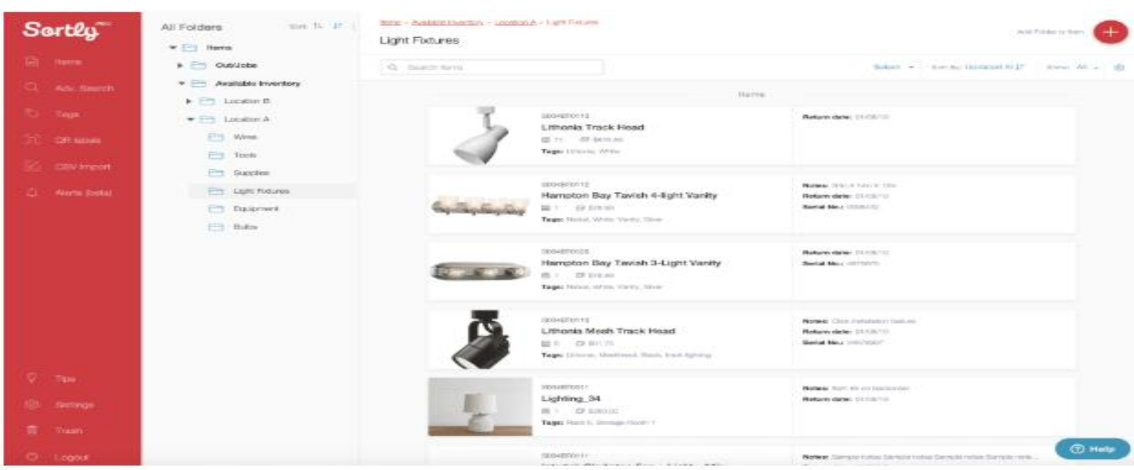



Figure 4 : **<u>Sortly Pro</u>**

Figure 4 shows the light furniture category.

**ZhenHub**

ZhenHub is a cloud-based logistics and inventory management solution for small and midsize businesses. In free version it offers tracking of the inventory, tracking of the shipment, and managing of the warehouse. ZhenHub's unique selling point is its shipping management functionality that integrates with multiple shipping solutions such as DHL and FedEx. It lets you schedule, manage, and track orders from these providers. The free version lets you manage one warehouse and supports up to 50 online orders per month (Srivastava, 2019).

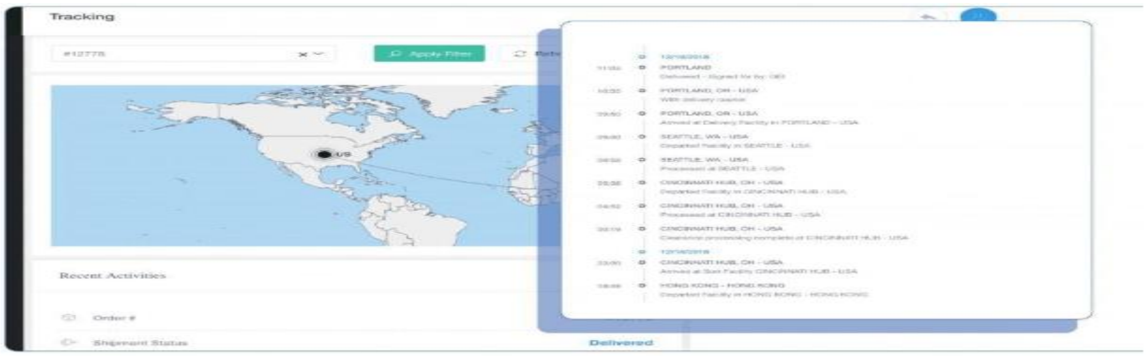



Figure 5 : **ZhenHub**

Figure 5 shows the shipment tracking.

**Zoho Inventory**

Zoho Inventory is a cloud-based inventory and warehouse management solution for small and midsize businesses.. Its free version enables to manage twenty online orders, twenty offline orders, twelve shipments, and 1 warehouse per month. This version additionally enables to choose and manage shipping suppliers for orders. Workflow management functionality is the free version's USP. It triggers associate alert as soon as the stock dips below the critical level and allows re-ordering of the stock (Srivastava, 2019).

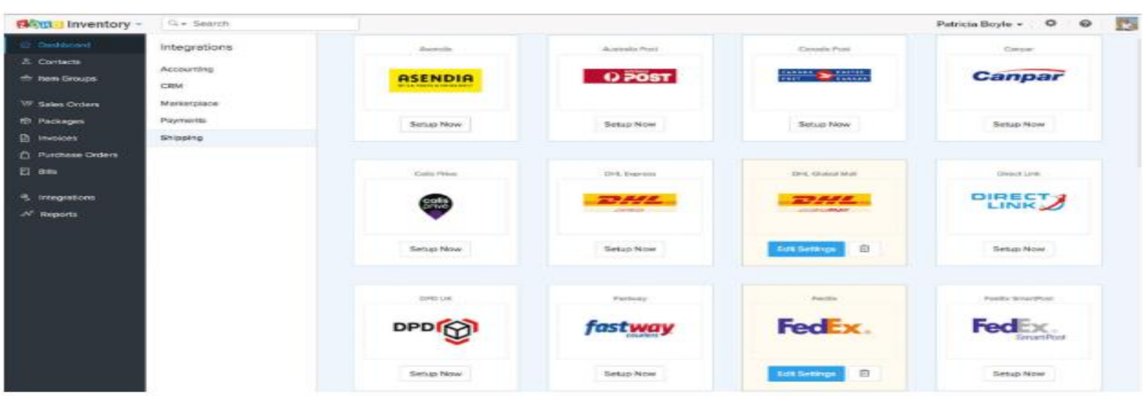



Figure 6 : **Zoho Inventory**

Figure 6 shows the twelve following shipping company that Zoho Inventory provides.

## Comparative Matrix

The illustration below shows the comparison between the six projects which are the ClickU: Kitchen Equipment Circulation and Inventory System, inFlow Inventory, Odoo, Sortly Pro, ZhenHub, and Zoho Inventory based by the five features sited.

Table 1

COMPARATIVE MATRIX

| | ClickU | inFlow Inventory | Odoo | Sortly Pro | ZhenHub | Zoho Inventory |
|---|---|---|---|---|---|---|
| Multi-user | ✓ | ✓ | × | ✓ | ✓ | ✓ |
| Notification | ✓ | × | × | × | × | ✓ |
| Track user activity/History Log | ✓ | × | × | × | × | × |
| Generate report | ✓ | ✓ | × | × | × | × |
| Restocking / Reorder | ✓ | × | × | × | × | ✓ |
| Reservation | ✓ | × | × | × | × | × |
| Inventory Management | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Admin can view Transaction | ✓ | × | × | × | × | × |
| User can view history | ✓ | × | × | × | × | × |
| Free | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

Table 1 illustrates that the project ClickU: Kitchen Equipment Circulation and Inventory System has all the eight features sited among the six projects. ClickU: Kitchen Equipment Circulation and Inventory System manage the kitchen inventory with efficiency and productivity in operations and manual tasks are done automated. Multi-user feature allows the user to access the system simultaneously. Notification feature reminds the user about unreturned items. Track user activity would monitor the user's borrowed utensils knowing whether it is returned yet or not and other transaction history. Generating a report would be much less of a hassle since it is just one click away. Users can also reserve kitchen equipment in ahead of time with the reservation feature. The admin can view transactions features, allows the admin to monitor and track logs. User can view history to check the utensils they have borrowed. All of these features make ClickU hassle-free, fast and reliable.

# CHAPTER III

# RESEARCH METHODOLOGY

This chapter presents the research methods applied in the development of the ClickU: Kitchen Equipment Circulation and Inventory System. The problems analyzed will be answered in this section.

## Research Environment

Research Environment is where the researchers conduct their research. It shows below the map or location of University of Cebu Lapu-Lapu and Mandaue.


Figure 7: **Location Map**

The study was conducted in the school of University of Cebu, the said location is chosen because the researchers are currently residing in College of Computer Science (CCS) Students, which enables them to conduct, gather and verify information necessary for the completion of the said study.

## Research Respondents

The researchers surveyed random respondents inside the school campus and those who are using the kitchen and other laboratories under HM Department. These respondents serve as the major source of data in developing the needed functionalities of the system. Fifty respondents were involved in the survey.

## Research Instruments

The researchers used a questionnaire that contains specific questions for gathering the necessary information relevant to the development of the system. The questions were based on facts, opinions, and observations that the researchers have encountered personally.

## Software Engineering Methodology

Software Development Methodology is a procedure in software engineering that is used in making the design, proposing and managing the process of developing software system.

Agile Software Methodology is the method to be used for the development process of a web-based system. It counts on the need for flexibility and implements a level of advantage into the implementation of the finished product. It emphasizes on making code understandable, frequent testing, and delivering functional parts of the application as soon as they are ready for deployment. The objective of Agile Software Methodology is to build upon small client-approved parts as the development progresses, as it contradicts to delivering the whole application at the end of the development phase.

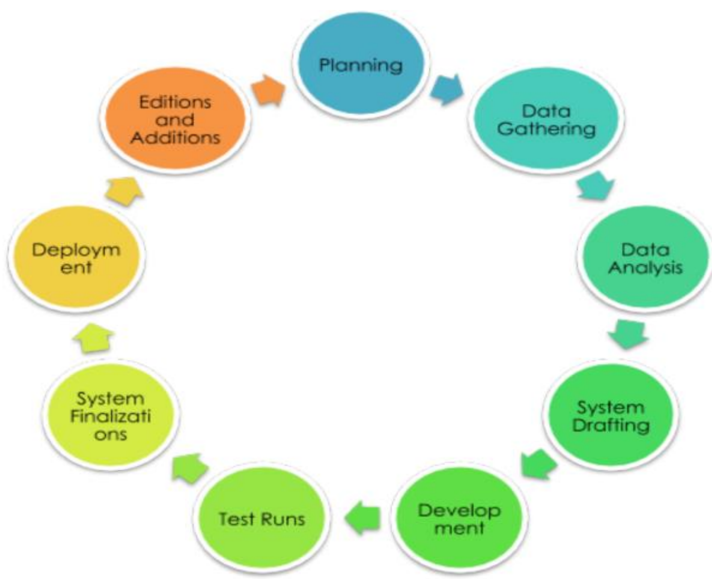



Figure 8: **Agile Development**

Figure 8 shows the nine phases of Agile Development on how to develop a web-based system.

## Evaluation of the Existing Laboratory Equipment Circulation System

Before planning for a computerized solution, the researchers conducted an evaluating of the existing system that is currently in place in the Hospitality Management System to further understand the process and identify that could help design for a software solution. The researchers conducted a survey and interview to students and faculty using a self-made survery questionnaire. In the interview and survey, the following problems were identified.

**Table 1**

**Evaluation on Borrowing Procedures**

| CRITERIA | WEIGHTED MEAN | VERBAL DESCRIPTION |
|---|---|---|
| Simple. | 2.61 | Agree |
| Does not require more time to process. | 2.45 | Disagree |
| Systematic. | 2.60 | Disagree |
| Provides information on available equipment quickly. | 2.61 | Agree |
| Does not require more information. | 2.58 | Disagree |
| Keeps records safely. | 2.53 | Disagree |
| Provides relevant information quickly. | 2.49 | Disagree |
| Provides accurate reports | 2.60 | Disagree |
| Provides on-time information | 2.68 | Agree |

**Table 2**

**Evaluation on Returning Procedures**

| CRITERIA | WEIGHTED MEAN | VERBAL DESCRIPTION |
|---|---|---|
| Simple. | 2.55 | Disagree |
| Does not require more time to process. | 2.54 | Disagree |
| Systematic. | 2.51 | Disagree |
| Provides information on available equipment quickly. | 2.35 | Disagree |
| Does not require more information. | 2.51 | Disagree |
| Keeps records safely. | 2.58 | Disagree |
| Provides relevant information quickly. | 2.55 | Disagree |
| Provides accurate reports | 2.40 | Disagree |
| Provides on-time information | 2.49 | Disagree |

Table 1 and 2 showed several problems in the borrowing and returning of equipment processes. Most of the problems were focused more on the procedures, data keeping and retrieval. These can be describes as follows;

1. The borrowing procedures is not simple and systematic.
2. The required information is not available on time.
3. The possibility of keeping erroneous data.
4. The records are not kept safe.

**Table 3**

**Evaluation on Equipment Inventory Procedures**

| CRITERIA | WEIGHTED MEAN | VERBAL DESCRIPTION |
|---|---|---|
| Simple and clear. | 2.55 | Disagree |
| Time-saving. | 2.50 | Disagree |
| Manually done. | 2.45 | Disagree |
| Accurate. | 2.35 | Disagree |
| Systematic | 2.25 | Disagree |
| Error prone | 2.50 | Disagree |

Table 3 showed that there is a low evaluation on the existing inventory procedures used in the Hospitality Management Department. The evaluation implies that there is a problem in the procedure, accuracy of data, productions of data takes time and a possibility to generate erroneous data.

**Table 4**

**Evaluation on Report Procedures**

| CRITERIA | WEIGHTED MEAN | VERBAL DESCRIPTION |
|---|---|---|
| Are up-to-date and on-time. | 2.40 | Disagree |
| Are accurate. | 2.55 | Disagree |
| Does not need more time to wait. | 2.60 | Disagree |
| Brief and concise. | 2.35 | Disagree |
| Has graphical representation. | 2.55 | Disagree |
| Are important for my decision-making and planning. | 2.60 | Disagree |
| Helps improve my job performance. | 2.55 | Disagree |
| Simplifies my job. | 2.60 | Disagree |

Table 4 showed that reports in not timely. The information generated from the existing system is not helpful in the decision-making. The existing system is not capable of generating graphs.

Based on the problems identified, the development of the software solutions would be focused on the following:

1. Simplicity and systematic procedures on the borrowing, returning and inventory processes.
2. Data could be generated accurately and timely.
3. The use of graphs should be integrated in aid for understanding the generated data which somehow important in the decision-making.
4. Data security should be taken into consideration.

## Planning/Conceptual-Initiation Phase

Planning and Initiation Phase involves the starting up of the project. This phase identifies the different functionalities, scopes, purposes, and outputs to be produced based on the evaluation of the existing system as seen in the result of table 1, 2 and 3. In the following phases emphasize more on the development of the software solution.

### Business Model Canvas

The following figure is the Business Model Canvas for ClickU: Kitchen Equipment Circulation and Inventory System. A web-based system, this system can help the people in the environment and in promoting the integration of modern technology to an otherwise mediocre system.

<table>
<tr>
<td colspan="2" rowspan="2">Key Partners<br><br>HM Department<br>School</td>
<td colspan="2">Key Activities<br><br>Software Development<br>Monitor and Manage Data</td>
<td colspan="2" rowspan="2">Value Proposition<br><br>ClickU is a Web-based Kitchen Equipment Circulation Inventory System for paperless transaction in borrowing of utensils</td>
<td colspan="2">Customer Relationship<br><br>Automated Self-service Personal</td>
<td colspan="2" rowspan="2">Customer Segments<br><br>Persons under school premises like students and guest</td>
</tr>
<tr>
<td colspan="2">Key Resources<br><br>Internet<br>Electricity<br>Partnership</td>
<td colspan="2">Channels<br><br>ClickU.com</td>
</tr>
<tr>
<td colspan="5">Cost Structure<br><br>Software Development Cost</td>
<td colspan="5">Revenue Streams<br><br>Free</td>
</tr>
</table>

Figure 9: **Business Model Canvas of ClickU**

Figure 9 shows the following key partners, key activities, value proposition, customer segments, customer relationship, cost structure, revenue streams, channels and key resources of the system ClickU.

**Program Workflow**

Activity Diagram as shown below is a flowchart to represent the flow from of one activity to another activity. The activity will be represented as an associate degree operation of the system. So the management flow is drawn from one operation to a different. This flow can be sequential, branched or concurrent.

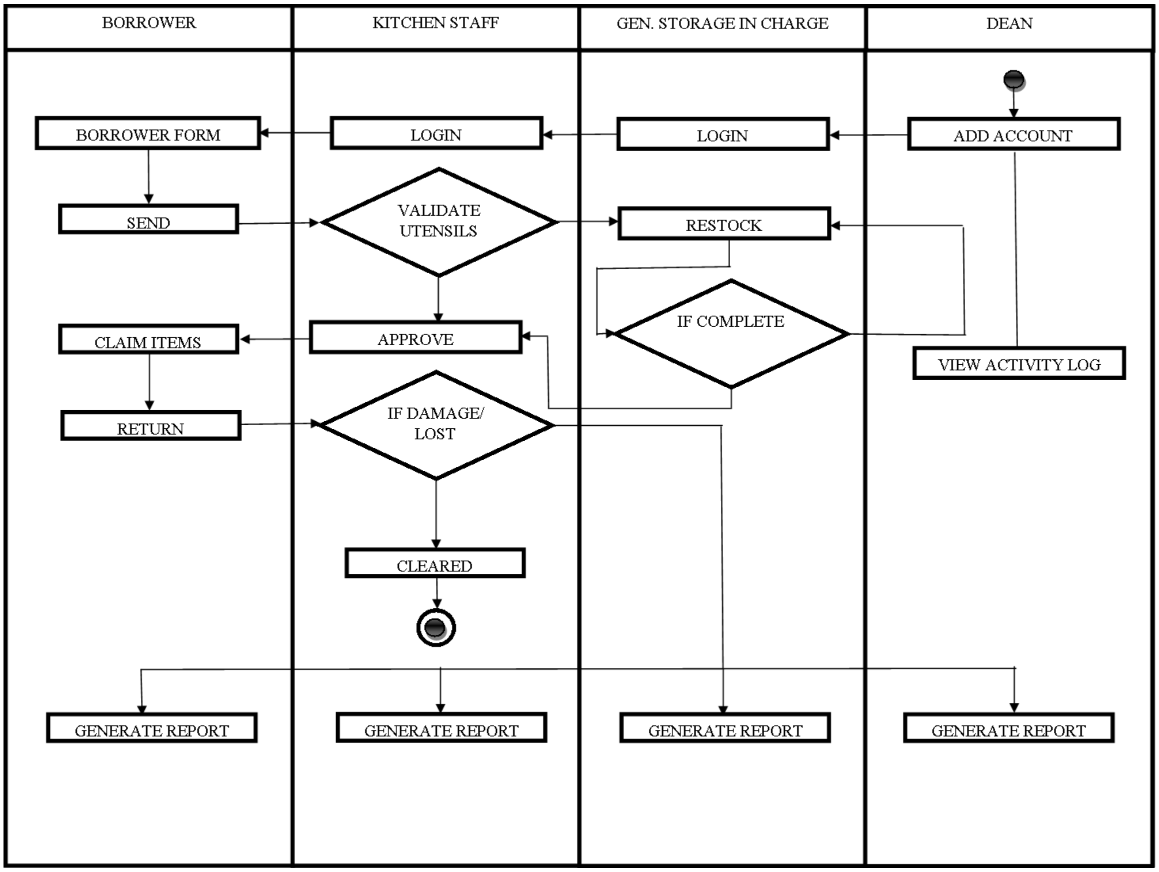



Figure 10: **<u>Activity Diagram</u>**

Figure 10 mainly shows the flow of ClickU web-based system in borrowing and returning of the kitchen equipment. The borrower completes the electronic borrower's form and then submit. The kitchen staff confirms the borrower's items once a borrower is identified. In the event of shortage of stock/utensils, the kitchen staff will borrow from the general storage which will be approved by the General Storage Incharge. The return of borrowed items will be approved once it has no damage or lost. The lost or damage items will be replaced by the borrower as part of the clearance requirements. The dean could also monitor the activity log of the users.

**Validation Board**

Validation Board is the best tool that is useful in helping the entrepreneurs in taking action while implementing the Lean Startup Process.

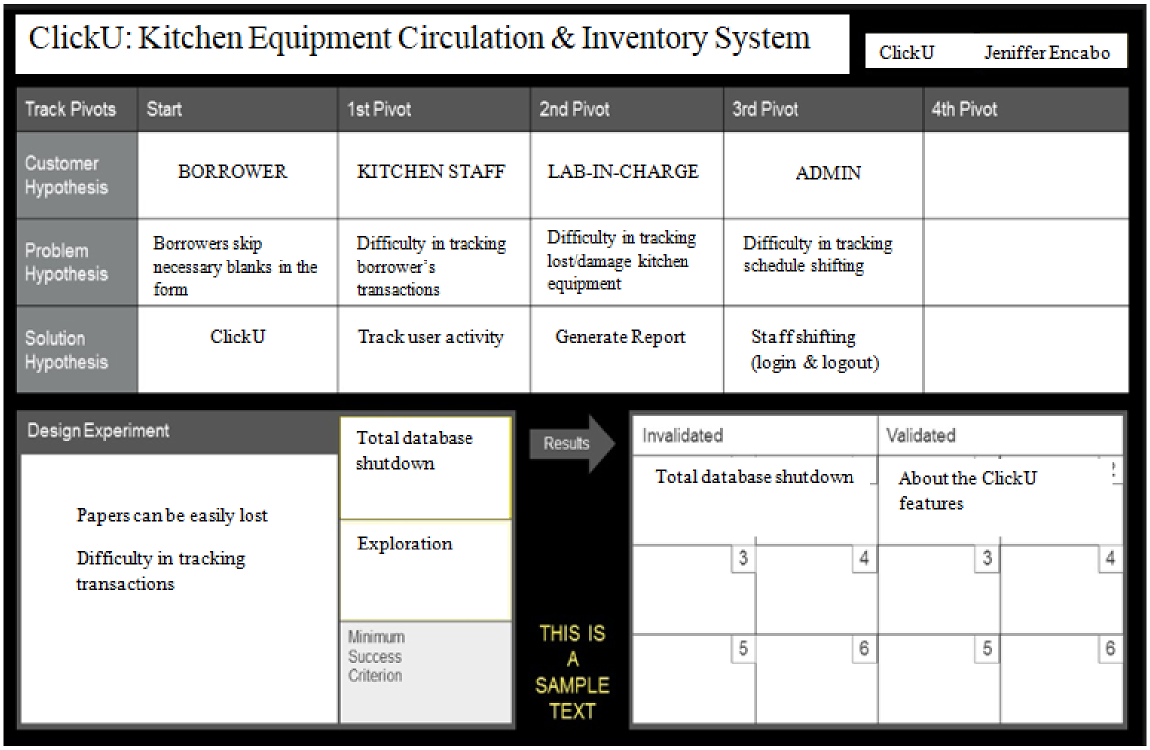



Figure 11: **<u>Validation Board of ClickU</u>**

Figure 11 shows the following information that will help the researchers in taking an action while implementing the ClickU web-based system.

**Gantt Chart**

The Gantt Chart illustrates the start and end dates of the different tasks assigned and a series of horizontal lines shows the progress of the work in a certain period with the allocated.

| Task ID | Task Name | Task Lead | Start Date | End Date |
|---|---|---|---|---|
| 1 | Planning | | | |
| 1.1 | Group Charter | All members | Jun. 19 | Jun. 22 |
| 1.2 | Group Meeting and Brainstorming | All members | Jun. 24 | Jul. 6 |
| 1.3 | Designing Business Model Canvas | All members | Jul. 8 | Jul. 13 |
| 1.4 | Designing Comparative Matrix | All members | Jul. 15 | Jul. 20 |
| 1.5 | Documentation | All members | Jun. 19 | Jul. 20 |
| 2 | Data Gathering Phase | | | |
| 2.1 | Conducting Survey | All members | Jul. 22 | Aug. 3 |
| 2.2 | Conducting Interview | All members | Aug. 5 | Aug. 10 |
| 2.3 | Documentation | All members | Sept. 9 | Aug. 3 |
| 3 | Data Analysis Phase | | | |
| 3.1 | Defining Problem | All members | Aug. 5 | Aug. 10 |
| 3.2 | Defining Objectives, Scope and Limitation | Hustler | Aug. 12 | Aug. 17 |
| 3.3 | Data Interpretation | Hustler | Aug. 19 | Aug. 24 |
| 3.4 | Creating Database Design | Hacker | Aug. 26 | Aug. 31 |
| 3.5 | Creating Database, ERD and Data Dictionary | Hacker | Sept. 2 | Sept. 7 |
| 3.6 | Documentation | All members | Aug. 5 | Sept. 7 |
| 4 | System Drafting Phase | | | |
| 4.1 | Defining System Functionalities | All members | Sept. 9 | Sept. 14 |
| 4.2 | Designing Business Use Case | Hacker | Sept. 16 | Sept. 21 |
| 4.3 | Program Workflow | Hacker | Sept. 16 | Sept. 21 |
| 4.4 | Designing Functional Decomposition Diagram | Hacker | Sept.23 | Sept. 28 |
| 4.5 | Designing Storyboard | Hacker | Sept.23 | Sept. 28 |
| 4.6 | Documentation | All members | Sept. 9 | Sept. 28 |
| 5 | Development Phase | | | |
| 5.1 | UX Development | Hipster | Nov. 11 | Nov. 20 |
| 5.2 | Web Coding | Hacker | Nov. 21 | Nov. 30 |
| 5.3 | Documentation | All members | Nov. 11 | Nov. 30 |
| 6 | Test Run Phase | | | |
| 6.1 | Creating Appropriate Tool | Tester | Dec. 2 | Dec. 11 |
| 6.2 | Program Debugging | Tester | Dec. 12 | Dec. 21 |
| 6.3 | Documentation | Hustler | Jan. 6 | Jan. 18 |
| 6.4 | Evaluating of Program Functionalities | All members | Jan. 20 | Feb. 1 |
| 6.5 | Documentation | All members | Dec. 2 | Feb. 1 |
| 7 | System Finalization Phase | | | |
| 7.1 | Finalizing the Design | All members | Feb. 3 | Feb. 15 |
| 7.2 | Checking Overall Structure of the System | All members | Feb. 17 | Feb. 29 |
| 9.1 | Documentation | All members | Feb. 3 | Feb. 29 |
| 8 | Deployment Phase | | | |
| 8.1 | Deploy device | All members | Feb. 3 | Mar. 21 |
| 8.2 | Final deployment of the Device | All members | Feb. 3 | Mar. 21 |
| 9.1 | Documentation | All members | Feb. 3 | Mar. 21 |
| 9 | Edition and Addition Phase | | | |

Legend:

| | |
|---|---|
| Done | |
| Progress | |
| Target | |
| Total | |

Figure 12: **Gantt Chart**

Figure 12 shows the period allocated of start and end dates of the different task of each following proponents in making the ClickU system. The researchers made a detailed documentation in conducting every phase of the study.

**Functional Decomposition Diagram**

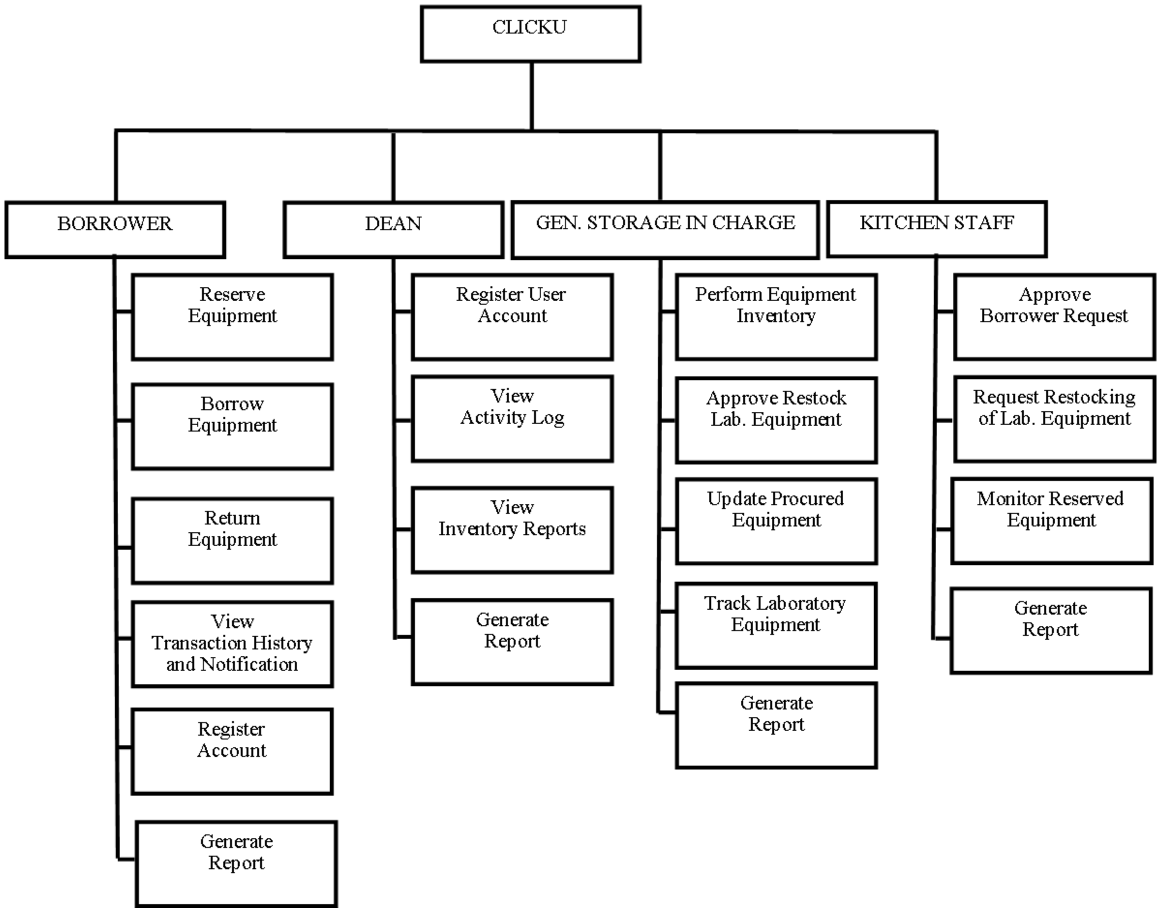



Figure 13: **Functional Decomposition Diagram of ClickU**

Figure 13 is the Functional Decomposition Diagram. A method of business analysis that compares distinct functional relationships that operate on a complex business process. It emphasizes on how the overall functionality is developed and its communication between various segments. It also shows the breakdown of functionalities in each user of the system.

## Analysis-Design Phase

### Data Gathering Phase

Support the facts and the process of gathering data to the problem definition and the feasibility of the system. The researchers use survey questionnaire (Appendix C) to gather data in identifying problems occurred in the existing system. to the following students, faculty staffs, working students who used the kitchen laboratory.

### Data Analysis Phase

The data analysis phase focus in the context of the proposed computer solution. The researchers used the following analysis tool namely: Use case diagram in Figure 14, Database design in Figure 34 and Entity-Relationship Diagram in Figure 35. The computer solution is clearly illustrated in the screen designs in Figure 24.

### Use Case Diagram

Use Case Diagram is a procedure in System Analysis and Design in which it is used to gather system requirements, identify internal and external factors influencing the system, and show interaction among the requirements and actors.

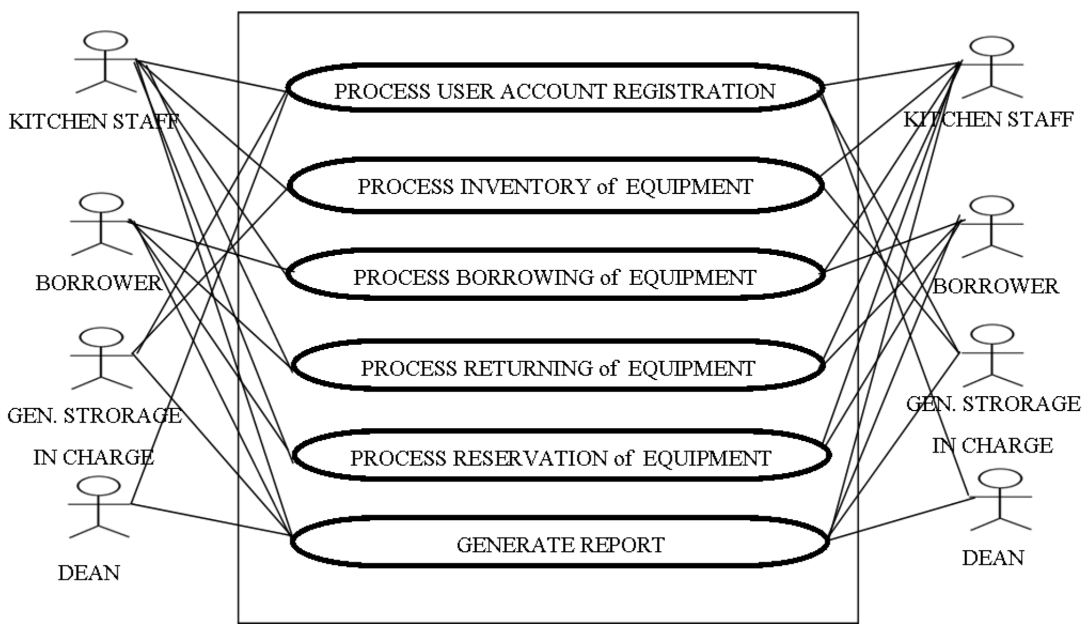



Figure 14: **Business Use Case**

Figure 14 shows the generalized main functionalities of the system.

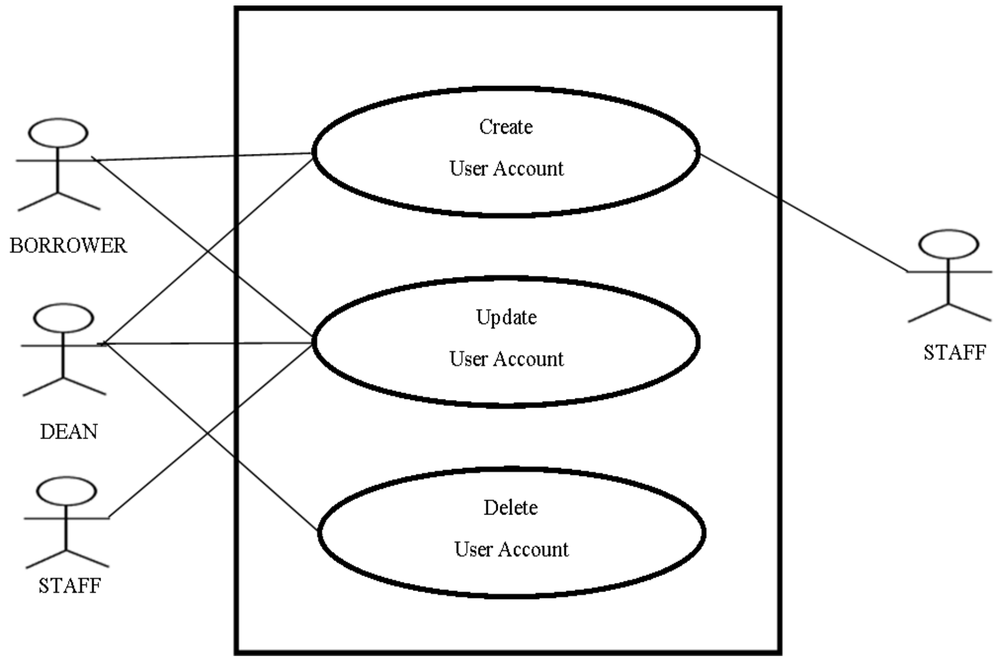



Figure 15: **Process User Account Registration Use Case**

**Name** : User Account Registration

**Purpose** : To allow the user to login in their account.

**Triggering** : Dean, Borrower & Staff

**Benefiting** : Staff

**Pre- Condition** : Input username and password.

**Post- Condition** : User account has been created.

**Steps** :

| USER | SYSTEM |
|---|---|
| 1. Input ID number<br>2. Input Password<br>3. Click Register button | 4. Validate Input<br>4.1. If not valid, an error message will prompt.<br>4.2. If valid, save to the database. |

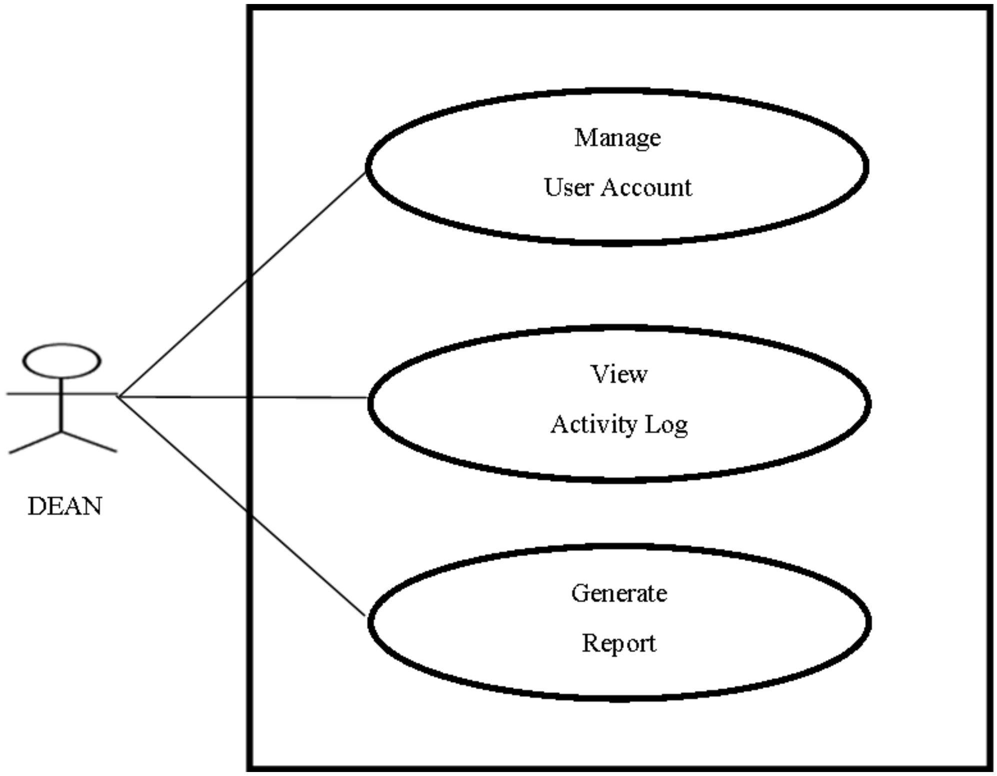



Figure 16: **Generate Report Use Case**

| | | |
|---|---|---|
| **Name** | : | Generate Report |
| **Purpose** | : | To allow the dean to check all the transaction logs and reports. |
| **Triggering** | : | Dean |
| **Benefiting** | : | System |
| **Pre- Condition** | : | View activity logs |
| **Post- Condition** | : | Generate reports |
| **Steps** | : | |

| DEAN | SYSTEM |
|---|---|
| 1. Click Accounts | 2. Redirect to Account page |
| 3. Click Activity Log button | 4. Redirect to Activity Log page |
| 5. Click Generate Reports button | 6. Generate Reports |

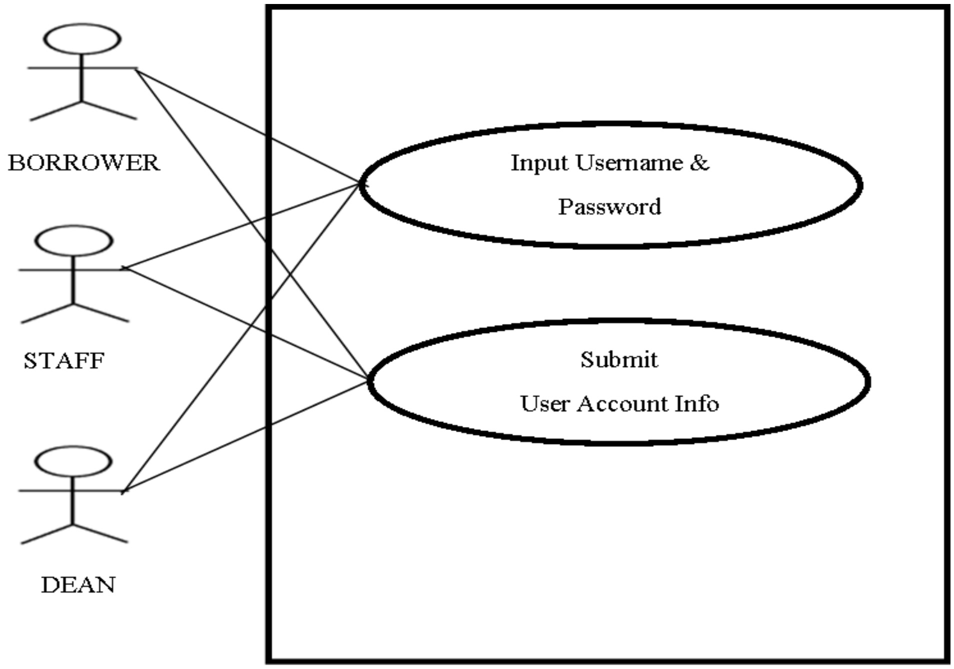



Figure 17: **Account Login Use Case**

**Name** : Account Login

**Purpose** : To allow the dean to manage account.

**Triggering** : Staff, Borrower & Dean

**Benefiting** : System

**Pre- Condition** : User account has been created.

**Post- Condition** : Account is successfully logged in.

**Steps** :

| USER | SYSTEM |
|---|---|
| 1. Click Login button<br>3. Input id number<br>4. Input password<br>5. Click Sign In button | 2. Redirect Login Form<br>6. Validate Input<br>6.1. If not valid, an error message will prompt.<br>6.2. If valid, redirect to home page |

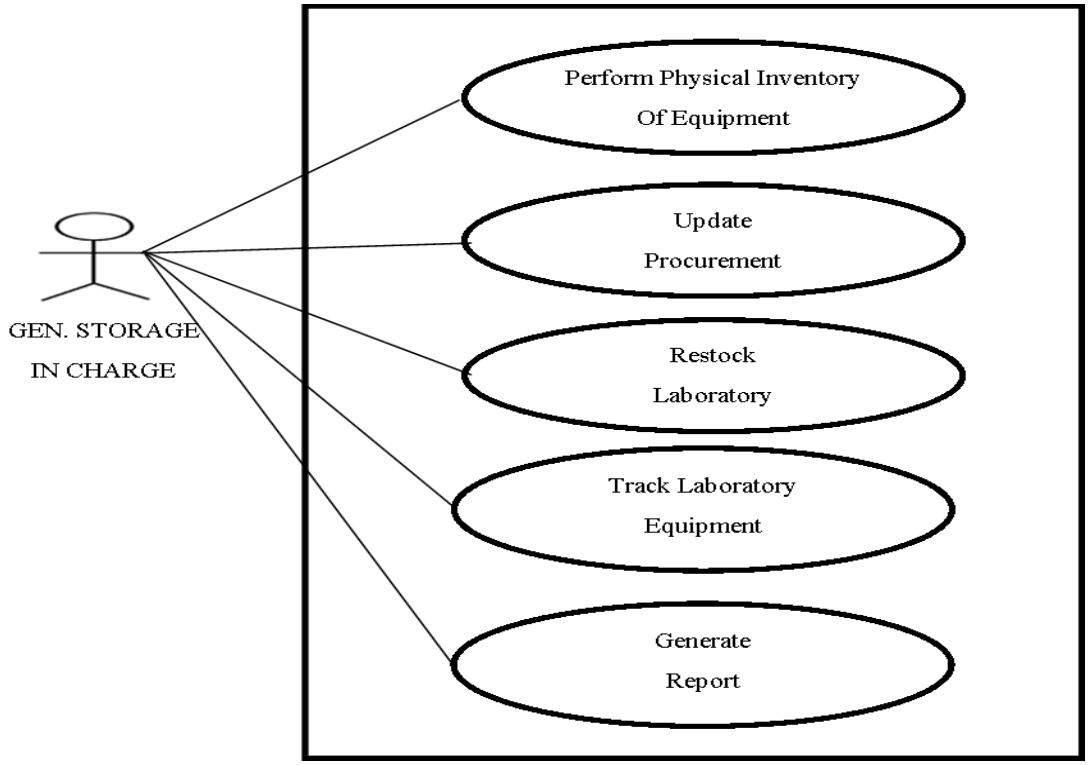



Figure 18: **<u>Process Inventory of Equipment Use Case</u>**

**Name** : Inventory of Equipment

**Purpose** : To allow the general storage in charge to monitor laboratory equipment.

**Triggering** : Gen. Storage In Charge

**Benefiting** : System

**Pre- Condition** : Validate the quantity of stock in the kitchen laboratory

**Post- Condition** : Kitchen laboratory stocks are completely stocked

**Steps** :

| GEN. STORAGE IN CHARGE | SYSTEM |
|---|---|
| 1. Manage Inventory<br>2. Restock Laboratory<br>4. Click Generate Report button | 3. Update quantity of kitchen equipment<br>5. Generate Report |

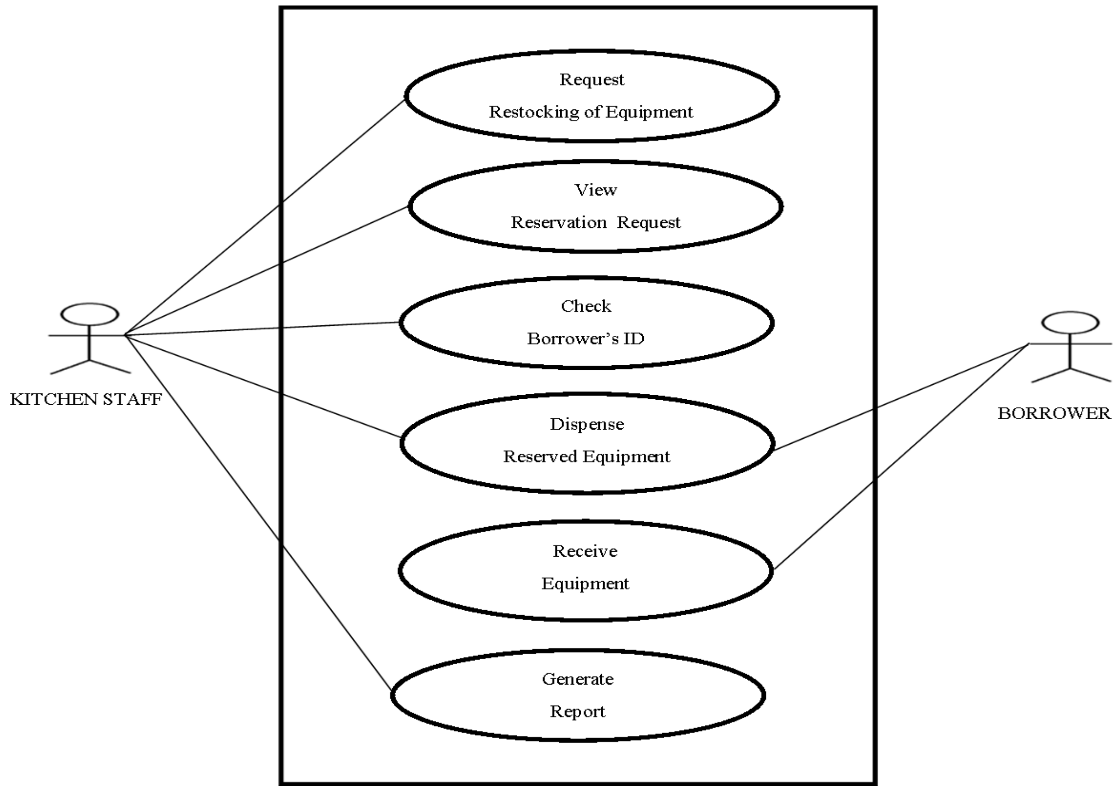



Figure 19: **Process Borrowing of Equipment Use Case**

**Name** : Borrowing of Equipment

**Purpose** : To allow the borrower to get the requested equipment

**Triggering** : Kitchen Staff

**Benefiting** : Borrower

**Pre- Condition** : Borrower submits reservation request.

**Post- Condition** : Kitchen Staff dispense kitchen equipment.

**Steps** :

| KITCHEN STAFF | SYSTEM |
|---|---|
| 1. Click Reservation Request<br>3. Click Generate Reports button | 2. Display details of Reservation Request<br>4. Generate Reports |

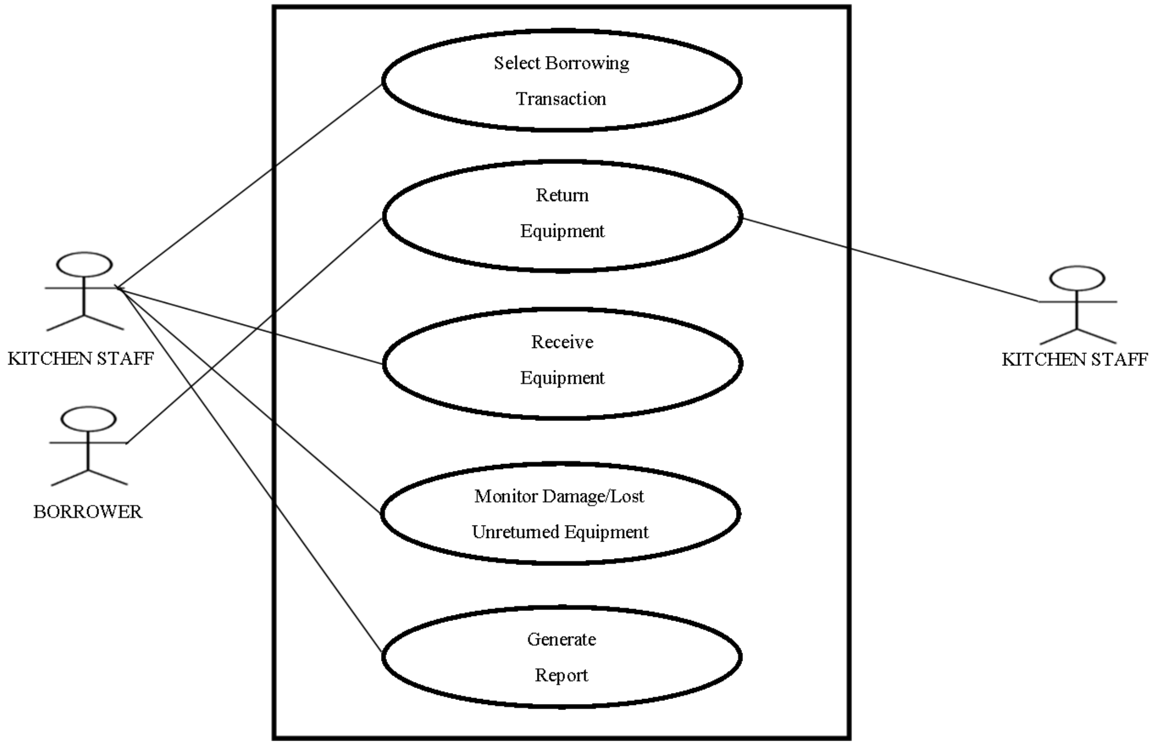



Figure 20: **<u>Process Returning of Equipment Use Case</u>**

**Name** : Returning of Equipment

**Purpose** : To allow the borrower to return the equipment used.

**Triggering** : Borrower

**Benefiting** : Kitchen Staff

**Pre- Condition** : Returns all equipment used.

**Post- Condition** : Successfully return all equipment used.

**Steps** :

| KITCHEN STAFF | SYSTEM |
|---|---|
| 1. Select Borrowing Transaction<br>3. Click Generate Reports button | 2. Display details of Borrowing Transaction<br>4. Generate Reports |

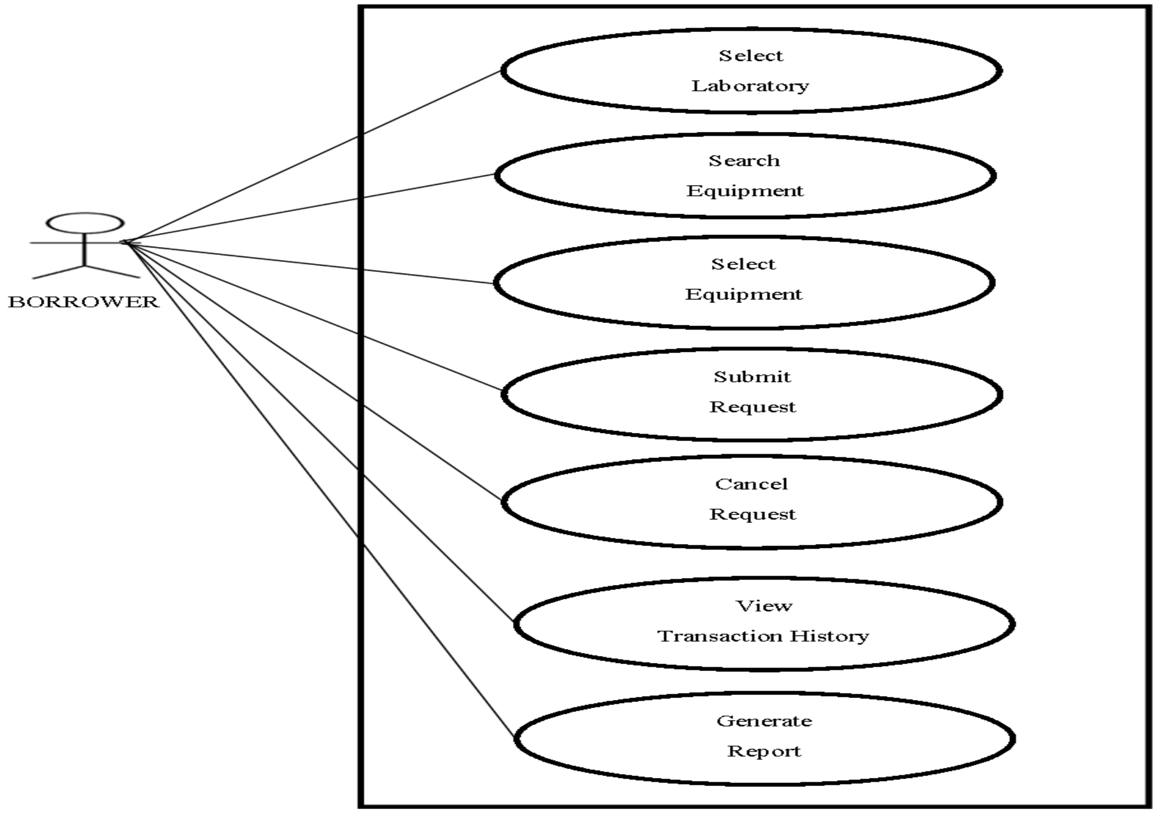



Figure 21: **Process Reservation of Equipment Use Case**

**Name** : Reservation of Equipment

**Purpose** : To allow the borrower reserve kitchen equipment ahead of time

**Triggering** : Borrower

**Benefiting** : System

**Pre- Condition** : Borrower selects equipment to reserve

**Post- Condition** : Borrower submits reservation request

**Steps** :

| BORROWER | SYSTEM |
|---|---|
| 1. Login account<br>3. Select Laboratory<br>4. Select Equipment<br>5. Click Reserve button | 2. Redirect to Reservation Form |

## System Drafting Phase

The start of the development process, this is the part where the researchers create the primary blueprint of the system and the structure of the data basis. The hacker of the team temporarily creates the blueprint of the database and user interface of the said system.

## Storyboard

Storyboard is a great way to visually depict an experience a reciprocal action between the user and the system. It consists of sequences of how the user responds to the system.

**MODULE NAME** **:** Main Homepage

**SCREEN NO.** **:** 1

**SCREEN NAME** **:** Homepage

**DESCRIPTION** **:** It is the home page where they can view the recent events posted in the department.


Figure 22: **Home Page Interface**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| Get Started | Button | Yes | | |
| Portfolio | Link | No | | |
| Login | Link | Yes | | |

Logic:

1. When the button "**GET STARTED**" is clicked, it will redirect to the borrower validation page.
2. When the link "**PORTFOLIO**" is clicked, it will display the department activity.
3. When the link "**LOGIN**" is clicked, it will redirect to the login page.

**MODULE NAME** : Account Login

**SCREEN NO.** : 2

**SCREEN NAME** : Account Login

**DESCRIPTION** : In login activity is where the users input ID number and password, and then clicks the sign in button. If correct, it will redirect to the homepage activity, otherwise prompted of the error.


Figure 23: **Account Login Interface**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| Username | Textbox | Yes | Varchar | 50 |
| Password | Textbox | Yes | Varchar | 50 |
| Sign in | Button | Yes | | |

Logic:

1. Fill up the log in form.

   1.1 If one of the fields is empty, an error message will be displayed.

   1.2 If users input a wrong username or password, an error message will be displayed.

2. When the “**Sign In**” button is clicked, it will redirect to the home page.

**MODULE NAME** : Account Registration

**SCREEN NO.** : 3

**SCREEN NAME** : Account Registration

**DESCRIPTION** : In registration activity is where the borrower will fill up registration form, and then clicks the register button.

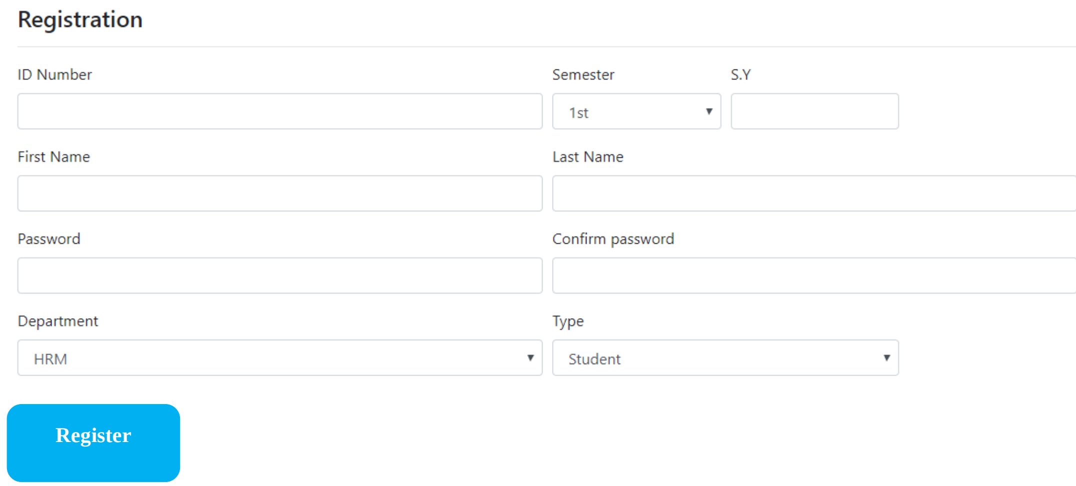



Figure 24: <u>**Account Registration Interface**</u>

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| ID number | Textbox | Yes | Integer | 11 |
| Semester | Dropdown | Yes | | |
| School year | Textbox | Yes | Integer | 11 |
| First name | Textbox | Yes | Varchar | 50 |
| Last name | Textbox | Yes | Varchar | 50 |
| Password | Textbox | Yes | Varchar | 50 |
| Department | Dropdown | Yes | | |
| Type | Dropdown | Yes | | |
| Register | Button | Yes | | |

Logic:

1. Fill up the registration form.

   1.1 If one of the fields is empty, an error message will be displayed.

   1.2 If password is not equal to confirm password, an error message will be displayed.

2. When the "**Register**" button is clicked, it will redirect to the login form.

**MODULE NAME** : Reservation of Kitchen Equipment

**SCREEN NO.** : 4

**SCREEN NAME** : Reservation of Kitchen Equipment

**DESCRIPTION** : In this activity is where the borrower can see the list of kitchen equipment that is available to be reserved and can select the desired equipment to reserve.

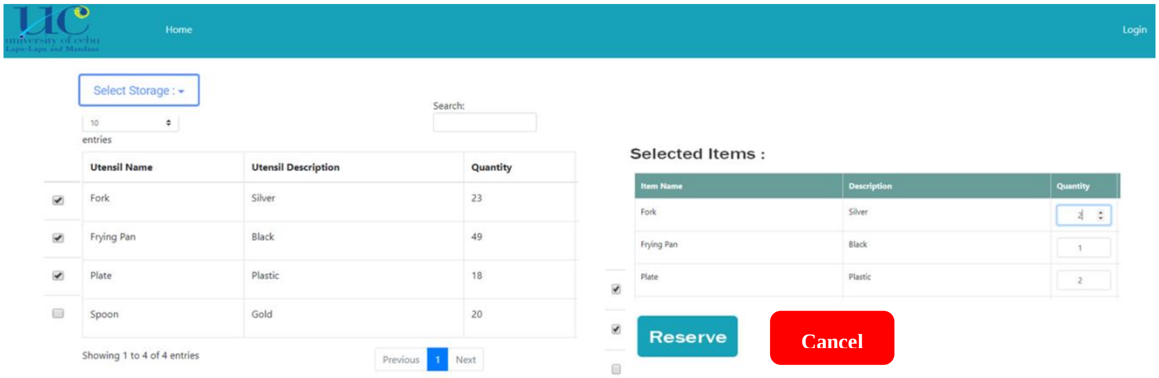



Figure 25: **<u>Reservation of Kitchen Equipment User Interface</u>**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| Storage | Dropdown | Yes | | |
| Select | Check Box | Yes | | |
| Search | Search Box | No | | |
| Reserve | Button | Yes | | |
| Quantity | Textbox | Yes | Integer | 11 |
| Cancel | Button | No | | |
| Previous | Button | Yes | | |
| Next | Button | Yes | | |

Logic:

1. Selects kitchen laboratory
2. Selects kitchen equipment to reserve.
3. When the button “Reserve” is clicked it will display at the reservation request.

**MODULE NAME** : List of Reservation Request

**SCREEN NO.** : 5

**SCREEN NAME** : List of Reservation Request

**DESCRIPTION** : In this activity it shows the list of reservation request that has to been submitted by the borrower.

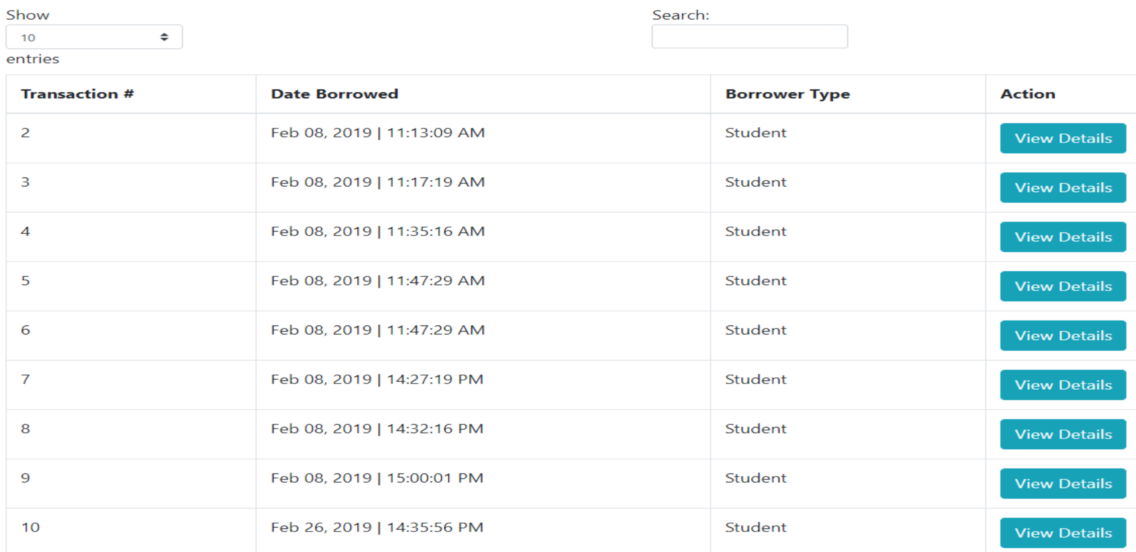



Figure 26: **List of Reservation Request Interface**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| View Details | Button | Yes | | |
| Search | Search Box | No | | |

Logic:

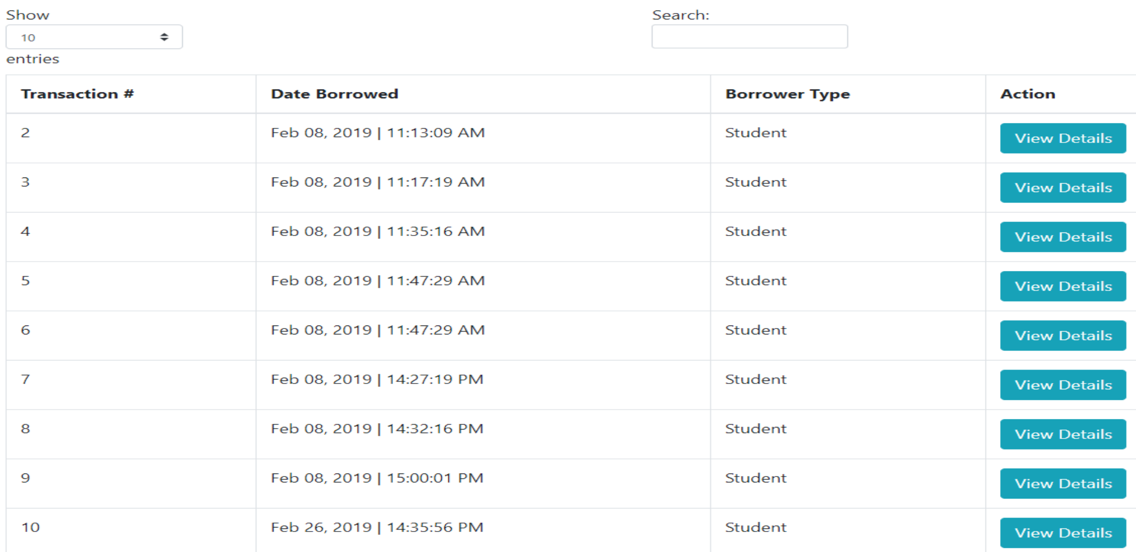
1. When the button “View Details” is clicked, it will display the reservation request details.

**MODULE NAME** : Returning of Equipment

**SCREEN NO.** : 6

**SCREEN NAME** : Returning of Equipment

**DESCRIPTION** : In this activity it shows the list of borrowing transaction that has the details of borrowed equipment.


Show 10 entries Search:

| Transaction # | Date Borrowed | Borrower Type | Action |
| --- | --- | --- | --- |
| 2 | Feb 08, 2019 \| 11:13:09 AM | Student | View Details |
| 3 | Feb 08, 2019 \| 11:17:19 AM | Student | View Details |
| 4 | Feb 08, 2019 \| 11:35:16 AM | Student | View Details |
| 5 | Feb 08, 2019 \| 11:47:29 AM | Student | View Details |
| 6 | Feb 08, 2019 \| 11:47:29 AM | Student | View Details |
| 7 | Feb 08, 2019 \| 14:27:19 PM | Student | View Details |
| 8 | Feb 08, 2019 \| 14:32:16 PM | Student | View Details |
| 9 | Feb 08, 2019 \| 15:00:01 PM | Student | View Details |
| 10 | Feb 26, 2019 \| 14:35:56 PM | Student | View Details |

Figure 27: **Returning of Equipment Interface**

| Items | Type | Required? | Data Type | Size |
| --- | --- | --- | --- | --- |
| View Details | Button | Yes | | |
| Search | Search Box | No | | |

Logic:

1. When the button “View Details” is clicked, it will display the borrowed equipment details

**MODULE NAME** : Remarks of Returned Equipment

**SCREEN NO.** : 7

**SCREEN NAME** : Remarks of Returned Equipment

**DESCRIPTION** : In this activity, happens the selecting of remarks of the following kitchen equipment that has been returned, lost or damage and unreturned.


Figure 28: **Remarks of Returned Equipment Interface**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| Approve | Button | Yes | | |
| Remark | Check Box | No | | |
| Quantity | Textbox | No | Integer | 11 |
| Status | Dropdown | No | | |

Logic:

1. Click checkbox for remark

   1.1 If the returned kitchen equipment has discrepancies

2. Select status
3. Input quantity of discrepancies
4. Click “approved” button

**MODULE NAME** : Adding of Kitchen Equipment

**SCREEN NO.** : 8

**SCREEN NAME** : Adding of Kitchen Equipment

**DESCRIPTION** : In this activity adding of the kitchen equipment occurs.

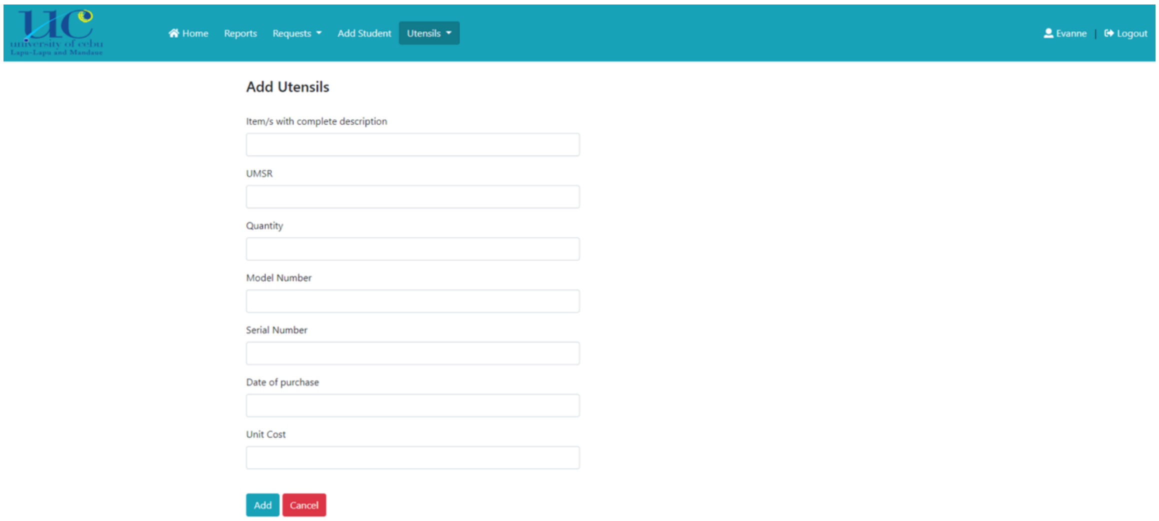



Figure 29: **Kitchen Equipment Data Entry Interface**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| Description | Textbox | Yes | Varchar | 50 |
| UMSR | Textbox | Yes | | |
| Quantity | Textbox | Yes | Integer | 11 |
| Model Number | Textbox | Yes | Integer | 11 |
| Serial Number | Textbox | Yes | Varchar | 50 |
| Date of Purchase | Textbox | Yes | Date | |
| Unit Cost | Textbox | Yes | Decimal | 11,2 |

Logic:

1. Fill up the adding of equipment form.

   1.1 If one of the fields is empty, an error message will be displayed.

2. When the button "**Add**" is clicked, it will be save to the database.

**MODULE NAME** : Generate List of Kitchen Equipment

**SCREEN NO.** : 9

**SCREEN NAME** : Generate List of Kitchen Equipment

**DESCRIPTION** : In this activity is where the general storage in charge can view the list of the kitchen equipment in every laboratory.

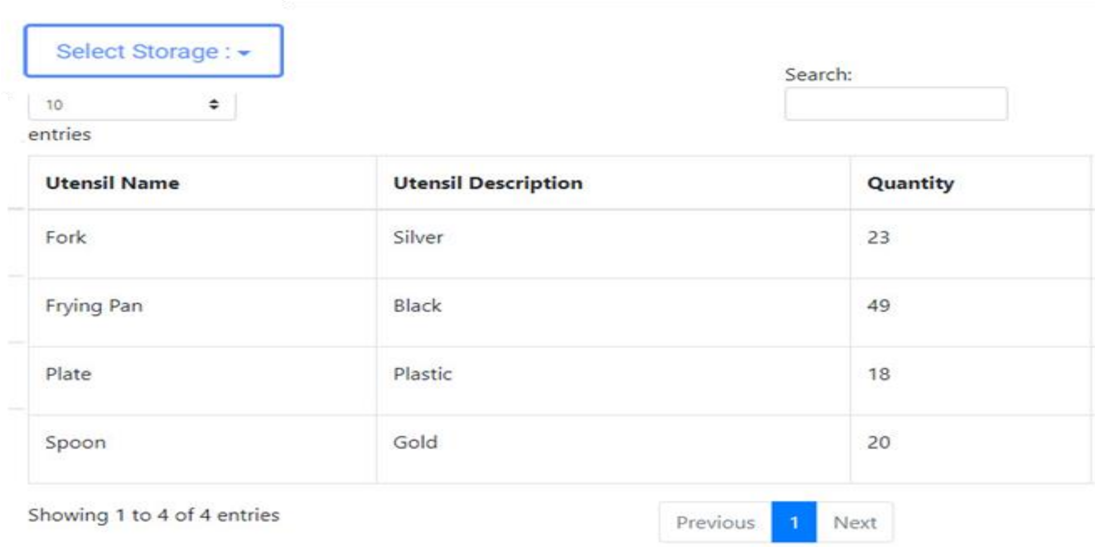



Figure 30: **Generate List of Kitchen Equipment Interface**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| Storage | Dropdown | Yes | | |
| Previous | Button | No | | |
| Next | Button | No | | |
| Search | Search Box | No | | |

Logic:

1. Select Laboratory.
2. Display the list of kitchen equipment.

**MODULE NAME** : Restocking of Equipment

**SCREEN NO.** : 10

**SCREEN NAME** : Restocking of Equipment

**DESCRIPTION** : In this activity is where the Gen. Storage in charge will add or restock the kitchen equipment for the the laboratory.

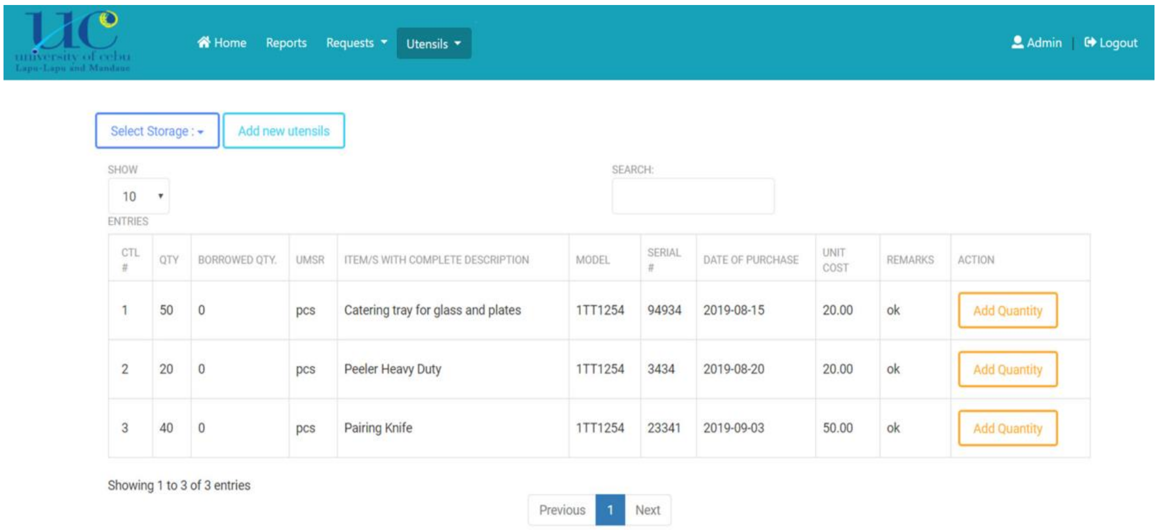



Figure 31: **<u>Restocking of Equipment Interface</u>**.

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| Select Storage | Button | Yes | | |
| Search | Search Box | No | | |
| Add new utensils | Button | No | | |
| Add Quantity | Button | Yes | | |
| Previous | Button | No | | |
| Next | Button | No | | |

Logic:

1. Click the “Select Storage” button to what laboratory will be restock
2. Click the “Add Quantity” to add quantity for that kind of equipment

**MODULE NAME** : Account Registration

**SCREEN NO.** : 11

**SCREEN NAME** : Account Registration

**DESCRIPTION** : In this activity is where the dean can create an account for kitchen staff and gen. storage in charge. Must fill in first name, last name, username, password, and select designation and type then clicks the Save button. An account will then be created.

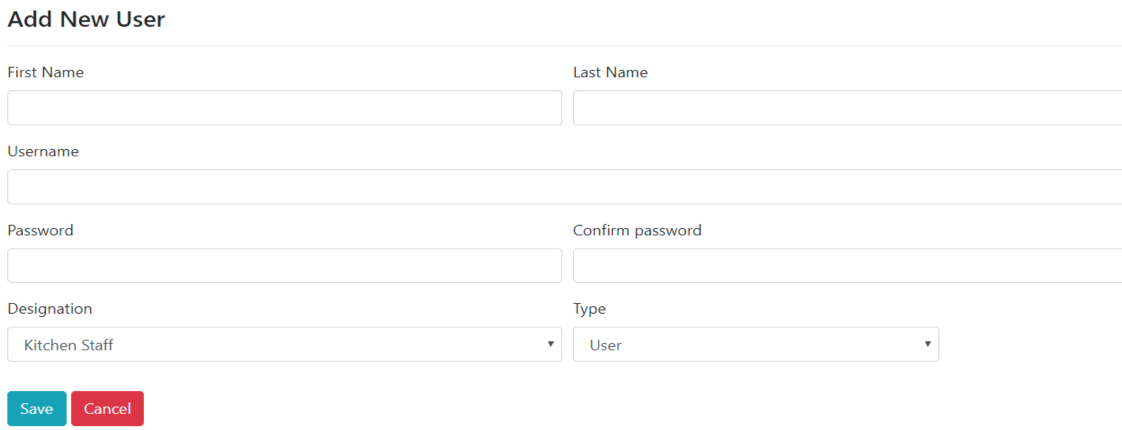



Figure 32: **<u>Registration of Staffs Interface</u>**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| First name | Textbox | Yes | Varchar | 50 |
| Last name | Textbox | Yes | Varchar | 50 |
| Username | Textbox | Yes | Varchar | 50 |
| Password | Textbox | Yes | Varchar | 50 |
| Designation | Dropdown | Yes | | |
| Type | Dropdown | Yes | | |
| Save | Button | Yes | | |
| Cancel | Button | No | | |

Logic:

1. 1. Fill up the registration form.

   1.1 If one of the fields is empty, an error message will be displayed.

   1.2 If the password is not equal to confirm password, an error message will be prompt.

   1.3 If the account already exists, an error message will be prompt.

2. When the button "**Save**" is clicked, it will be saved to the database.

**MODULE NAME** : Generate Report

**SCREEN NO.** : 12

**SCREEN NAME** : Generate Report

**DESCRIPTION** : In this activity is where the user can generate report respectively.

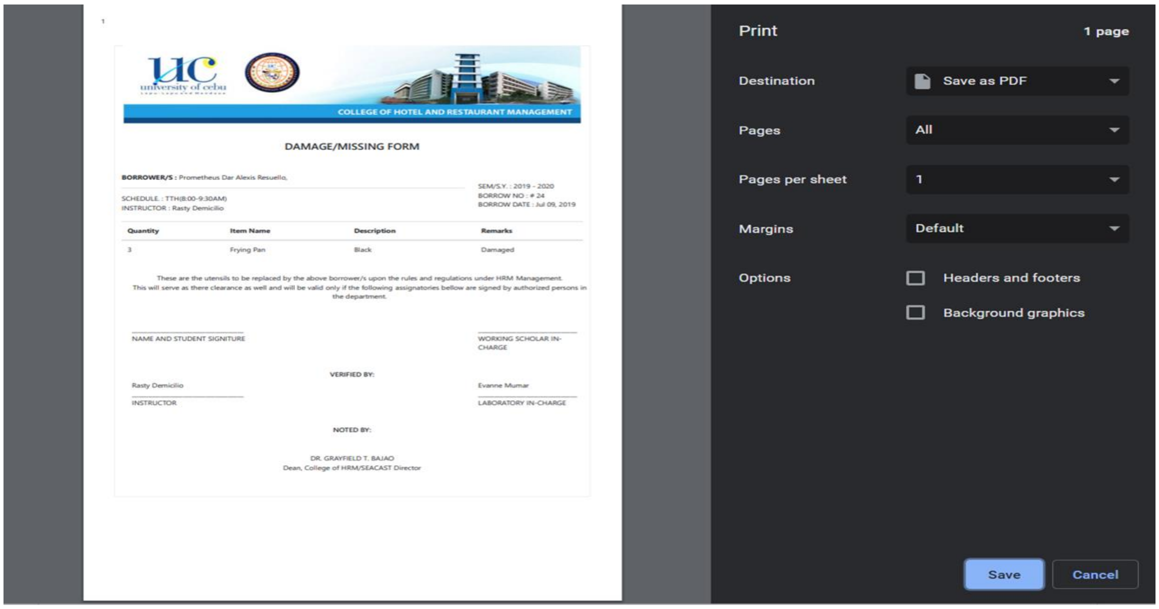



Figure 33: **<u>Generate Report Interface</u>**

| Items | Type | Required? | Data Type | Size |
|---|---|---|---|---|
| Destination | Dropdown | Yes | | |
| Pages | Dropdown | Yes | | |
| Pages per sheet | Dropdown | Yes | | |
| Margin | Dropdown | Yes | | |
| Options | Check box | Yes | | |
| Save | Button | Yes | | |
| Cancel | Button | No | | |

Logic:

1. Generates Report
2. When the button "Save" is clicked, it will be save to its selected destination.

**Database Design**

Database Diagram is the organization of data according to a database model. The designer determines what information should be keep and the way the info components interrelate. With this info, they will begin to suit the info to the info model.

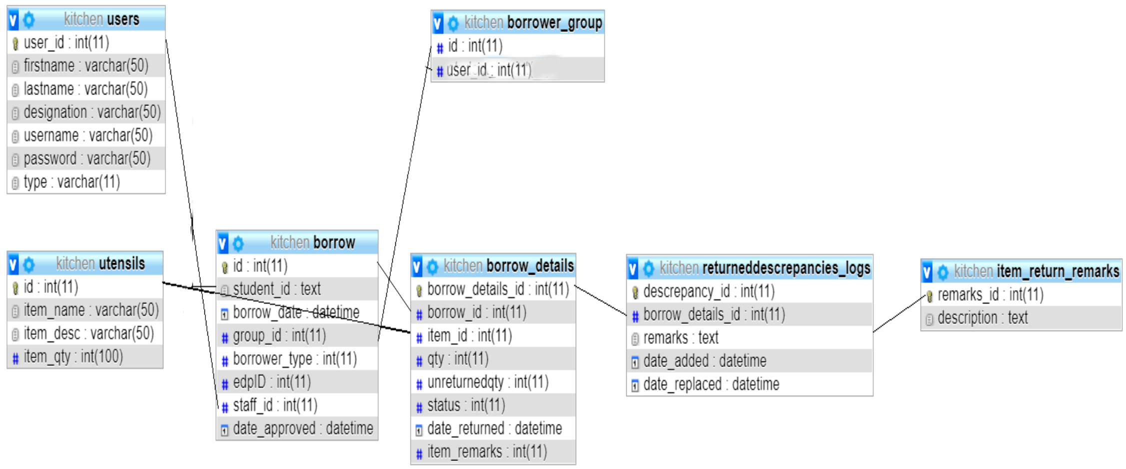



Figure 34: **Database Design of ClickU**

Figure 34 shows the tables used for the system.

**Entity-Relationship Diagram**

An ERD is an abstract and realistic model of knowledge accustomed to represent the entity framework infrastructure.

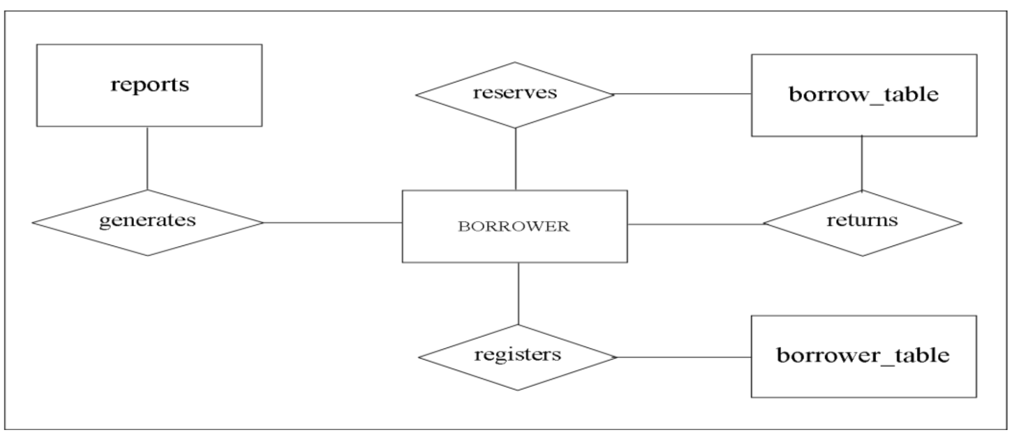



Figure 35: **Entity-Relationship Diagram**

Figure 35 shows the entity framework infrastructure for the borrower's side.

**Data Dictionary**

A data dictionary could be a file or a collection of files that contains a database's data. The data dictionary contains records concerning to different objects within the database, such as data ownership, data relationships to other objects and other data.

Table 2

BORROW TABLE

| NAME | DATA TYPE | KEY | NOT NULL/NULL | DESCRIPTION |
|---|---|---|---|---|
| id | int(11) | PK | NOT NULL | Transaction Number |
| student_id | text | FK | NOT NULL | Student ID |
| borrow_date | datetime | | NOT NULL | Date Borrow |
| group_id | int(11) | FK | NOT NULL | Group ID |
| borrower_type | int(11) | | NOT NULL | Borrower Type |
| edpID | int(11) | | NOT NULL | EDP Code ID |
| staff_id | int(11) | | NOT NULL | Staff ID |
| data_approved | datetime | | NOT NULL | Date Approved |

Table 2 shows to be the master table for borrowers with the borrow date included and the id will serve as the transaction number.

Table 3

BORROWER GROUP TABLE

| NAME | DATA TYPE | KEY | NOT NULL/NULL | DESCRIPTION |
|---|---|---|---|---|
| Id | int(11) | FK | NOT NULL | Individual and Group ID |
| student_id | int(11) | FK | NOT NULL | Student ID |

Table 3 shows that the data for group of borrowers will be stored here by using the id and borrower_id as foreign keys to link with the borrow table.

Table 4

BORROW DETAIL TABLE

| NAME | DATA TYPE | KEY | NOT NULL/NULL | DESCRIPTION |
|---|---|---|---|---|
| borrow_detail_id | int(11) | PK | NOT NULL | Borrow detail ID |
| borrow_id | int(11) | FK | NOT NULL | Individual and Group ID |
| item_id | int(11) | FK | NOT NULL | Item ID |
| qty | int(11) | | NOT NULL | Item Quantity |
| unreturnedqty | int(11) | | NULL | Unreturned Item Quantity |
| status | int(11) | | NOT NULL | Approval Status |
| date_returned | datetime | | NOT NULL | Date Returned |
| item_remarks | int(11) | | NOT NULL | Remarks |

Table 4 shows it holds the details of the borrowed items such as the quantity of unreturned items, the date and the status.

Table 5

RETURN REMARKS TABLE

| NAME | DATA TYPE | KEY | NOT NULL/NULL | DESCRIPTION |
|---|---|---|---|---|
| remarks_id | int(11) | PK | NOT NULL | Remarks ID |
| description | text | | NOT NULL | Description |

Table 5 shows where the remarks of the unreturned items will be placed.

Table 6

RETURNED DESCREPANCIES TABLE

| NAME | DATA TYPE | KEY | NOT NULL/NULL | DESCRIPTION |
|---|---|---|---|---|
| discrepancy_id | int(11) | PK | NOT NULL | Discrepancy ID |
| borrow_details_id | int(11) | FK | NOT NULL | Borrow detail ID |
| remarks | text | FK | NOT NULL | Remarks |
| date_added | datetime | | NOT NULL | Date Added |
| date_replaced | datetime | | NOT NULL | Date Replaced |

Table 6 shows the data of the items with discrepancies will be stored in this table.

Table 7

USERS TABLE

| NAME | DATA TYPE | KEY | NOT NULL/NULL | DESCRIPTION |
|---|---|---|---|---|
| user_id | int(11) | PK | NOT NULL | Staff and Admin ID |
| firstname | varchar(50) | | NOT NULL | First Name |
| lastname | varchar(50) | | NOT NULL | Last Name |
| designation | varchar(50) | | NOT NULL | Designation |
| username | varchar(50) | | NOT NULL | Username |
| password | varchar(50) | | NOT NULL | Password |
| type | int(11) | | NOT NULL | User Type |

Table 7 shows where the data of the users will be stored and will be used for validation.

Table 8

UTENSILS TABLE

| NAME | DATA TYPE | KEY | NOT NULL/NULL | DESCRIPTION |
|---|---|---|---|---|
| id | int(11) | PK | NOT NULL | Utensil ID |
| item_name | varchar(50) | | NOT NULL | Utensil Name |
| item_desc | varchar(50) | | NOT NULL | Utensil Description |
| item_qty | int(100) | | NOT NULL | Utensil Quantity |

Table 8 shows the utensils and equipment data will be stored in this table.

**Network Design**

Network design involves evaluating, understanding, scoping, and planning of the implementation of a computer network infrastructure.

**Network Model**

The Network model is an information model that shows the relationships among the objects. The schema of network model is viewed as a graph with nodes and connecting links.

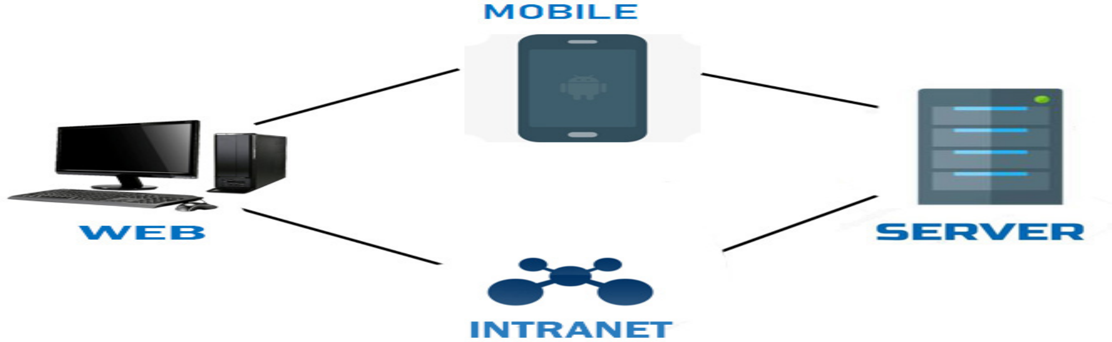



Figure 36: **Network Model**

Figure 36 shows the connecting relationship of the web, the intranet, the mobile and the server.

**Network Topology**

Network Topology refers to layout of a network. How completely different nodes during a network square measure connected to every different and the way they convey is set by the network's topology.

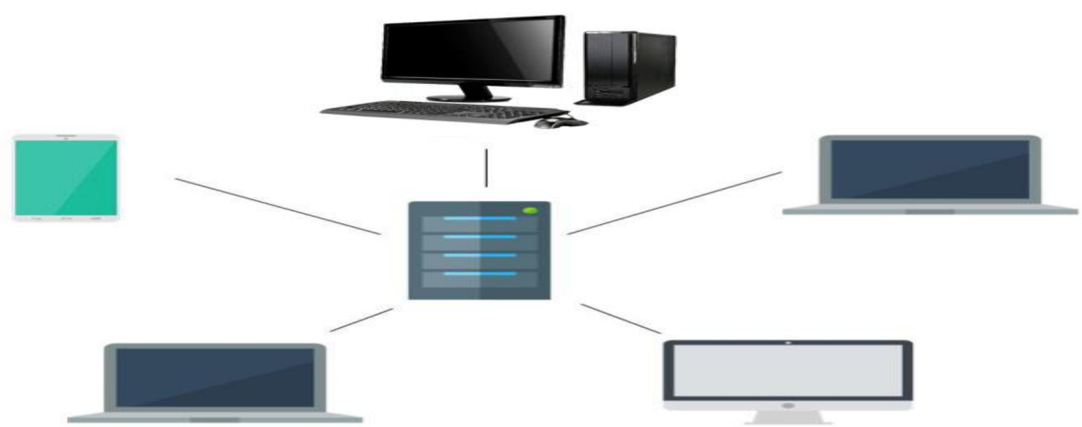


Figure 37: **Network Topology**

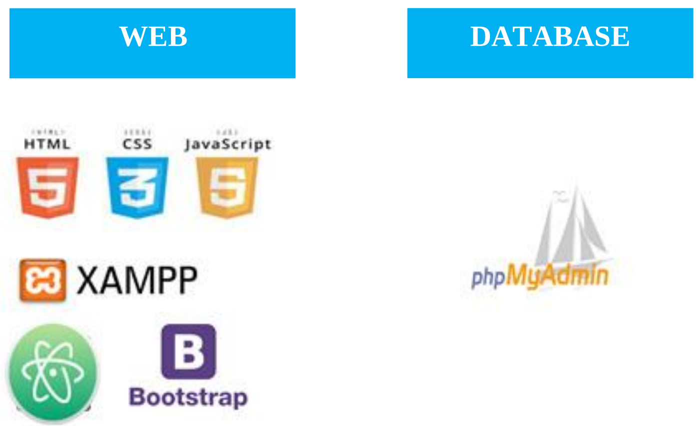
Figure 37 shows a layout of star topology of network

## Development/Construction/Build Phase

Development/Construction/Build Phase involves the development of the project. This phase contains the Software Specification and Hardware Specification which is under by the Technology Stack Diagram and List of Module under by the Program Specification.

## Technology Stack Diagram

Technology Stack is a product of hardware requirements, software products and programming languages used to create the ClickU: Kitchen Equipment Circulation and Inventory System (Web-based system).




## Software Specification

This section discusses the software specifications of ClickU: Kitchen Equipment Circulation and Inventory System (Web-based system). This includes the programming language, operating system version, the platform, framework and the database software that will be used for the deployment of the system. In developing the system, HTML, CCS, Java script and PHP will be used for the front end of the web-based system and phpMyAdmin will be used as the back-end or the database. Windows 7 (All versions); Windows 8 (All versions); Windows 10 (All builds) are the operating system that the researchers will use for the Web and PHP is the programming language that will use for web.

## Hardware Specification

This section discusses the software specifications of ClickU: Kitchen Equipment Circulation and Inventory System (Web-based system). This includes the web, central processing unit (Core i5 7300HQ @ 2.50GHz (4CPUs),~2.5GHz), random access memory (16gb(8x2) DDR4 2400MHz), motherboard (ASUSTeK COMPUTER INC. Intel Kaby Lake), monitor (ASUS 15.6”  IPS LED Monitor), and storage (Kingston A400 240GB SSD for Operating System and Western Digital 2TB HDD for data) in developing the system.

## Program Specification

In program design, a specification captures precisely the interfaces between the modules of a program. It contains the list of modules that the researchers are going to implement for the system.

## List of Modules

This section shows the assignments of each proponent in the team.

Table 9

LIST OF MODULES

| Programmer | Modules |
|---|---|
| **WEB** | |
| Meljun Tirol | Generate Reports |
| | Validate Borrower |
| | Remarks |
| Jeniffer Encabo | Login Page |
| | Registration Page |
| | Add Kitchen Equipment |
| John Carl Sanico | Home Page Layout |
| | Admin Home Page Layout |
| | List of Kitchen Equipment |
| Prometheus Dar Alexis Resuello | Home Page |
| | Approval |
| | View Transactions |

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# CURRICULUM VITAE

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| *2013 - 2014*<br>(*Undergraduate)* | **Cebu Technological University – Main Campus**<br>Cebu City, Cebu<br>Bachelor of Science in Mechatronics Major in Robotics |
| *2008 - 2012* | **Marigondon National High School**<br>Marigondon, Lapu-Lapu City, Cebu |
| *2003 - 2008* | **Alegria Elementary School**<br>Alegria, Cordova, Cebu |

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| *2011 – 2014* | **Mandaue City Comprehensive National High School**<br>Mandaue City, Cebu |
| *2002 – 2008* | **Cesar M. Cabahug Elementary School**<br>Looc, Mandaue City, Cebu |

# APPENDICES

## Appendix A

## Transmittal Letter

University of Cebu Lapu-Lapu and Mandaue
A.C. Cortes Ave., Looc, Mandaue City
College of Computer Studies

October 17, 2019

**DR. GRAYFIELD BAJAO**
Hospitality Management Department
Dean

Dear Dr. Bajao:

Good day.
We the 4th year Bachelor of Science in Information Technology (BSIT) students of University of Cebu Lapu-Lapu and Mandaue will be developing a web-based system called "ClickU: Kitchen Equipment Circulation and Inventory System" for HM Department. This is in response to your request. Therefore, we are asking for your approval to please allow us to conduct a survey and interview in order to obtain the requirements in implementing the system.

We are hoping for your positive response on this matter.
Thank you and God Bless!

Respectfully yours,

**JENIFFER ENCABO**
Project Manager

Noted by

**ENGR. VIRGINIA B. VERDUN**
Adviser

Approved by

**DR. AURORA C. MIRO**
CCS, Dean

**Appendix B**

**Research Environment**

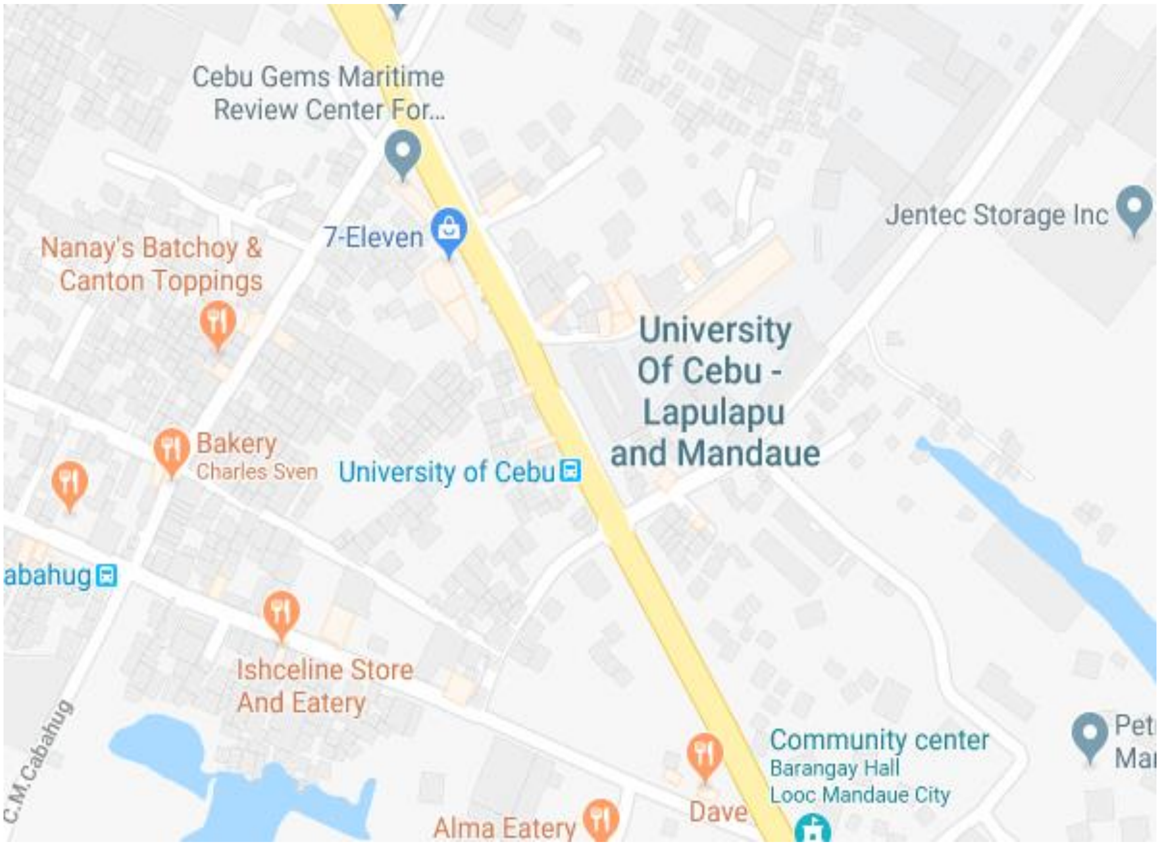


Figure: **Location Map**

**Appendix C**

**Research Survey Instrument**

**SURVEY QUESTIONNAIRE**

(for Students)

**PART I. PROFILING OF THE RESPONDENTS**

Name: _________________

Year level: ______________

**PART II. ASSESSMENT OF THE EXISTING PROCEDURES FOR LABORATORY EQUIPMENT MONITORING**

**Direction:** This questionnaire is intended for the computerization of the laboratory equipment circulation and inventory system of HM Department. The researcher would like you to give a general evaluation of the existing system by checking on the column that corresponds to your evaluation.

SA Strongly agree
MD Moderately agree
A Agree
D Disagree
SD Strongly Disagree

| INDICATORS | SA (5) | MD (4) | A (3) | D (2) | SD (1) |
|---|---|---|---|---|---|
| **Borrowing procedures:** | | | | | |
| 1. Simple. | | | | | |
| 2. Does not require more time to process. | | | | | |
| 3. Systematic. | | | | | |
| 4. Provides information on available equipment quickly. | | | | | |
| 5. Does not require more information. | | | | | |
| 6. Keeps records safely. | | | | | |
| 7. Provides relevant information quickly. | | | | | |
| 8. Provides accurate reports | | | | | |
| 9. Provides on-time information | | | | | |
| **Returning procedures:** | | | | | |
| 1. Simple. | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| 2. Does not require more time to process. | | | | | |
| 3. Systematic. | | | | | |
| 4. Provides information on available equipment quickly. | | | | | |
| 5. Does not require more information. | | | | | |
| 6. Keeps records safely. | | | | | |
| 7. Provides relevant information quickly. | | | | | |
| 8. Provides accurate reports | | | | | |
| 9. Provides on-time information | | | | | |

**SURVEY QUESTIONNAIRE**

(for Faculty)

**PART I. PROFILING OF THE RESPONDENTS**

Name: __________________

Designation: _______________

**PART II. ASSESSMENT OF THE EXISTING PROCEDURES FOR EQUIPMENT MONITORING**

**Direction:** This questionnaire is intended for the computerization of the laboratory equipment circulation and inventory system of HM Department. The researcher would like you to give a general evaluation of the existing system by checking on the column that corresponds to your evaluation.

SA Strongly agree
MD Moderately agree
A Agree
D Disagree
SD Strongly Disagree

| INDICATORS | SA (5) | MD (4) | A (3) | D (2) | SD (1) |
|---|---|---|---|---|---|
| **Equipment Inventory procedures:** | | | | | |
| 1. Simple and clear. | | | | | |
| 2. Time-saving. | | | | | |
| 3. Manually done. | | | | | |
| 4. Accurate. | | | | | |
| 5. Systematic | | | | | |
| 6. Error prone | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **The reports provided by the existing system:** | | | | | |
| 1. Are up-to-date and on-time. | | | | | |
| 2. Are accurate. | | | | | |
| 3. Does not need more time to wait. | | | | | |
| 4. Brief and concise. | | | | | |
| 5. Has graphical representation. | | | | | |
| 6. Are important for my decision-making and planning. | | | | | |
| 7. Helps improve my job performance. | | | | | |
| 8. Simplifies my job. | | | | | |
| **Equipment borrowing procedures:** | | | | | |
| 1. Simple. | | | | | |
| 2. Does not require more time to process. | | | | | |
| 3. Systematic. | | | | | |
| 4. Provides information on available equipment quickly. | | | | | |
| 5. Does not require more information. | | | | | |
| 6. Keeps records safely. | | | | | |
| 7. Provides relevant information quickly. | | | | | |
| 8. Provides accurate reports. | | | | | |
| 9. Provides on-time information. | | | | | |
| **Equipment returning procedures:** | | | | | |
| 1. Simple. | | | | | |
| 2. Does not require more time to process. | | | | | |
| 3. Systematic. | | | | | |
| 4. Provides information on available equipment quickly. | | | | | |
| 5. Does not require more information. | | | | | |
| 6. Keeps records safely. | | | | | |
| 7. Provides relevant information quickly. | | | | | |
| 8. Provides accurate reports | | | | | |
| 9. Provides on-time information | | | | | |